

Method of Double Moments for 3-D Reconstruction in Cryo-EM: Using Multiple Datasets with Different Viewing Angle Distributions

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Abstract

Cryo-electron microscopy (cryo-EM) is a widely used imaging technique for reconstructing three-dimensional molecular structures based on noisy and randomly oriented tomographic projection images. We introduce a new data fusion approach, termed the method of double moments, to reconstruct the molecular structure using two instances of the second-order moment of the projection images, which are generated from two different distributions of random orientations (e.g., arising from different experimental conditions), where one distribution is uniform and the other is non-uniform and unknown. We prove that these moments generically uniquely identify the structure, up to a global rotation and reflection. Furthermore, we devise a convex relaxation-based algorithm which numerically recovers bandlimited structures from these moments, demonstrating that reliable recovery is possible using only second-order statistics.

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1 Introduction

Cryo-electron microscopy has become an increasingly popular technique for single-particle three-dimensional structure reconstruction [23, 31, 3, 14]. It aims to recover the underlying molecular structure from a large collection of noisy 2-D tomographic projection images taken at unknown and typically random orientations, with applications in structural biology, medicine, and drug discovery [44, 57, 20, 49, 60].

A fundamental challenge in cryo-EM is the high noise level and the random viewing directions of particles in the sample, which complicate the reconstruction process. This paper introduces a new statistical approach for 3-D reconstruction that relies solely on low-order statistics—specifically, two instances of the second-order moment of the observed projection images, each computed from images with different distributions of orientations. Crucially, we prove that under mild conditions, these moments uniquely determine the underlying structure (up to a global rotation and reflection), and we devise a computationally efficient algorithm based on convex relaxation to perform the reconstruction.

The sample complexity of our method scales as $\omega(\text{SNR}^{-2})$, where SNR denotes the signal-to-noise ratio of the input images. This improves upon previous method-of-moments approaches that either require access to higher order moments, typically resulting in sample complexity scaling at least as $\omega(\text{SNR}^{-3})$, or do not offer uniqueness guarantees. Furthermore, in contrast to earlier methods that often suffer from spurious local minima and stagnation, our algorithm demonstrates robust performance in numerical experiments. See Section 1.3 for a detailed comparison with related work.

An important practical motivation for our method comes from the fact that projection orientations in cryo-EM are often not uniformly distributed. This anisotropy in orientation distribution can result from experimental factors such as preferred particle orientations or sample preparation artifacts [56, 37, 24, 42, 15, 38, 4]. To address this issue, recent experimental advances have sought to manipulate the orientation distribution to be more uniform—for example, by tilting the specimen [2], laser flash melting [55], adding charged detergents [36], or embedding molecules in liposomes [63].

Our proposed method, termed the Method of Double Moments (MoDM), is naturally suited to such experimental scenarios: it leverages two datasets collected under different orientation distributions—one ideally uniform, made possible by experimental interventions, and the other non-uniform and unknown a priori. While this setting is increasingly relevant in practice, it has not been fully explored in prior statistical reconstruction frameworks. Our results demonstrate that access to such complementary datasets enables exact structure recovery using only second-order statistics, thereby opening new avenues for robust and efficient cryo-EM reconstruction despite variability in particle orientations. We also remark that other imaging techniques, like X-Ray Free-Electron Lasers (XFEL), could benefit from incorporating datasets with different viewing distributions, and our approach could be applicable.

The remainder of this paper is organized as follows. Section 1.1 introduces the notation used throughout the paper. Section 1.2 describes the image formation model considered in cryo-EM, and Section 1.3 reviews related work. Section 2 details the assumptions and conven-

tions adopted in this study and precisely states the main reconstruction problem. Section 3 introduces the proposed algorithm for solving the problem. In particular, Section 3.5 provides numerical experiments showing the effectiveness of the method. Section 4 proves that the problem admits a unique solution. Finally, Section 5 concludes the paper. The code implementing the proposed algorithm is available at <https://github.com/oscar Mickelin/modm>.

1.1 Notation

Denote by $\mathbb{Z}_{\geq 0}$ the set of nonnegative integers, and by $\mathbf{1}_{\mathcal{X}}$ the indicator function of a set $\mathcal{X}$. For a matrix M , denote by $M^{\top}$ and $M^{\mathbf{H}}$ its transpose and conjugate transpose, respectively. We denote by $\mathbf{O}(n)$ the set of real orthogonal matrices of dimension n , that is, $\mathbf{O}(n) = \{M \in \mathbb{R}^{n \times n} : MM^{\top} = M^{\top}M = I\}$. Similarly, $\mathbf{SO}(n)$ denotes the special orthogonal group of dimension n , consisting of real orthogonal matrices with determinant $+1$, i.e., $\mathbf{SO}(n) = \{M \in \mathbf{O}(n) : \det(M) = 1\}$. We also use $\mathbf{O}(n, \mathbb{C})$ to denote the set of complex orthogonal matrices of dimension n , defined by $\mathbf{O}(n, \mathbb{C}) = \{M \in \mathbb{C}^{n \times n} : MM^{\top} = M^{\top}M = I\}$. Note that, unlike unitary matrices, the orthogonality condition here does not involve complex conjugation.

For matrices $M_1 \in \mathbb{C}^{n_1 \times n_1}, \dots, M_N \in \mathbb{C}^{n_N \times n_N}$, let $\text{blockdiag}_{i=1, \dots, N}(M_i)$ denote the block-diagonal matrix of size $(\sum_{i=1}^N n_i) \times (\sum_{i=1}^N n_i)$ defined as

$$\text{blockdiag}_{i=1, \dots, N}(M_i) := \begin{bmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_N \end{bmatrix}.$$

Equivalently, for two matrices M_1 and M_2 , we write $M_1 \oplus M_2$ to denote their block-diagonal concatenation:

$$M_1 \oplus M_2 := \begin{bmatrix} M_1 & 0 \\ 0 & M_2 \end{bmatrix}.$$

This binary operation can be extended recursively to N matrices as

$$M_1 \oplus M_2 \oplus \cdots \oplus M_N := \text{blockdiag}_{i=1, \dots, N}(M_i).$$

For integrable functions $f, g : \mathbb{R}^d \rightarrow \mathbb{C}$, the convolution of f and g is defined as

$$(f * g)(\mathbf{x}) := \int_{\mathbb{R}^d} f(\mathbf{x} - \mathbf{y}) g(\mathbf{y}) d\mathbf{y}, \quad \mathbf{x} \in \mathbb{R}^d.$$

For a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we use the following convention for the Fourier transform:

$$\hat{f}(\boldsymbol{\omega}) = \int_{\mathbb{R}^d} f(\mathbf{x}) e^{-2\pi i \mathbf{x} \cdot \boldsymbol{\omega}} d\mathbf{x},$$

where $\mathbf{x} \cdot \boldsymbol{\omega} = \sum_{j=1}^d x_j \omega_j$ denotes the Euclidean inner product.

1.2 Image formation model

We denote the electrostatic potential of a target 3-D molecular structure by $\Phi^* : \mathbb{R}^3 \rightarrow \mathbb{R}$, and represent a random rotation by $R \in \text{SO}(3)$. The action of R on Φ^* is written as $R^\top \cdot \Phi^* : \mathbb{R}^3 \rightarrow \mathbb{R}$, and defined by

$$(R^\top \cdot \Phi^*)(x_1, x_2, x_3) := \Phi^*(R(x_1, x_2, x_3)), \quad \text{for } (x_1, x_2, x_3) \in \mathbb{R}^3, \quad (1.1)$$

where the operator $R^\top \cdot \Phi^*$ denotes the rotated potential via a change of coordinates by the rotation R . The corresponding tomographic projection image $I_R(x_1, x_2)$, acquired under viewing direction R , is modeled as

$$I_R(x_1, x_2) = \int_{-\infty}^{\infty} (R^\top \cdot \Phi^*)(x_1, x_2, x_3) dx_3 + \varepsilon(x_1, x_2), \quad \text{for } (x_1, x_2) \in \mathbb{R}^2, \quad (1.2)$$

where ε is an additive Gaussian white noise term with variance $\sigma^2 I$, independent of the signal. The noise level σ^2 can typically be estimated in advance from the observed images. We further assume the random rotation R follows an unknown probability distribution ρ^* over the rotation group $\text{SO}(3)$. In a standard cryo-EM experiment, one observes a collection of noisy 2-D projection images $I_1, \dots, I_N$, where each image I_i is associated with an independent unknown rotation $R_i \sim \rho^*$.

To better understand the relationship between the projections and the 3-D structure, it is convenient to consider the 2-D Fourier transform of the projection images. By the Fourier slice theorem [41], the 2-D Fourier transform $\widehat{I}_R$ of a projection image I_R corresponds to a central planar slice of the 3-D Fourier transform $\widehat{\Phi^*}$ of the volume, taken perpendicular to the viewing orientation R . Specifically,

$$\widehat{I}_R(\omega_1, \omega_2) = (R^\top \cdot \widehat{\Phi^*})(\omega_1, \omega_2, 0) + \widehat{\varepsilon}(\omega_1, \omega_2), \quad \text{for } (\omega_1, \omega_2) \in \mathbb{R}^2, \quad (1.3)$$

This frequency-domain formulation plays a central role in many reconstruction algorithms and is particularly useful for statistical modeling of image formation under varying orientations.

The goal is to recover the structure Φ^* from the noisy projection images, without prior knowledge of the distribution ρ^* (see also Section 2.4). Due to the high noise level in individual images, achieving meaningful resolution in the reconstructed 3-D structure typically requires tens of thousands of projection images or more [3, 60].

A more refined and realistic image formation model also accounts for optical aberrations introduced by the microscope, modeled via convolution with a point spread function $h_i(x_1, x_2)$, whose Fourier transform is known as the contrast transfer function (CTF) [53]. Each h_i is typically a radially symmetric, highly oscillatory function with frequent zero crossings, which complicates the inversion process. Under this model, the projection image I_i corresponding to rotation R_i is given by

$$I_i(x_1, x_2) = h_i(x_1, x_2) * \int_{-\infty}^{\infty} (R_i^\top \cdot \Phi^*)(x_1, x_2, x_3) dx_3 + \varepsilon(x_1, x_2), \quad \text{for } (x_1, x_2) \in \mathbb{R}^2.$$

In the proposed algorithm, the projection images $\{I_i\}_{i=1}^N$ are used solely to estimate the second-order moment of the underlying structure. The effect of the point spread function can be corrected during this estimation, provided that the functions h_i , for $i = 1, \dots, N$, have sufficiently non-overlapping zero-crossings [50, 39]. We therefore assume that such a correction has been applied and omit the point spread functions in the subsequent analysis.

1.3 Existing methods

Maximum likelihood-based approaches. Maximum likelihood estimation provides a principled statistical framework for 3-D structure reconstruction. The goal is to estimate the unknown 3-D molecular structure by maximizing the likelihood of observing a given set of 2-D projection images $\{I_i\}_{i=1}^N$, as described by the formation model (1.2). This leads to the following objective function:

$$\mathcal{L}(\Phi) = \sum_{i=1}^N \log \int_{\text{SO}(3)} p(I_i|\Phi, R)p(R|\Phi) dR. \quad (1.4)$$

Here, $p(I_i|\Phi, R)$ denotes the likelihood of observing image I_i given a rotation R and the 3-D structure Φ , and $p(R|\Phi)$ represents a prior over rotations, typically assumed to be uniform due to the lack of knowledge about the true rotation distribution ρ^* . This formulation marginalizes out the latent rotation variable R , reflecting the fact that it is unobserved during data acquisition. Since this is a classical example of an incomplete data problem, the expectation-maximization (EM) algorithm is commonly employed to maximize the likelihood [17]. Popular state-of-the-art methods resort to EM-based procedures—commonly referred to as 3-D iterative refinement—which alternate between estimating the posterior distribution over rotations (E-step) and updating the 3-D structure estimate by maximizing the expected log-likelihood (M-step) [51, 46, 43].

Despite their empirical success, these approaches suffer from non-convexity, and no global convergence guarantees are known [52, 53]. As a result, the algorithm may converge to a local maximum, particularly if the initialization is not sufficiently close to the ground truth. This is problematic for downstream applications such as drug discovery and design, where reconstruction accuracy and reliability are critical [40, 44]. Moreover, the methods are computationally intensive, requiring access to the entire dataset during each iteration, further limiting scalability.

Method of moments. An alternative approach is based on the method of moments, using experimental data to compute empirical moments of the *Fourier transforms* of the projection images, at increasing orders. The k th-order empirical moment $\tilde{m}_k : \mathbb{R}^{2k} \rightarrow \mathbb{C}$ is given by¹

$$\tilde{m}_k(\omega_1, \dots, \omega_k) := \frac{1}{N} \sum_{i=1}^N \hat{I}_i(\omega_1) \cdots \hat{I}_i(\omega_{k-1}) \overline{\hat{I}_i(\omega_k)} - B_k(\omega_1, \dots, \omega_k, \sigma), \quad (1.5)$$

¹The complex conjugate in the last mode ensures symmetry properties of the second-order moment; some related work uses different conventions.

where $\omega_j \in \mathbb{R}^2$, for $j = 1, \dots, k$, indexes a location in the 2-D Fourier domain of the projected images, and should be understood as part of the coordinate system of the moment tensor. Here, $\widehat{I}_i$ denotes the 2-D Fourier transform of the i -th projection image. The term $B_k(\omega_1, \dots, \omega_k, \sigma)$ denotes a debias term that depends on the noise variance of the projection images (which we assume has been estimated from the data). Since $\widetilde{m}_k$ is computed from data and thus implicitly depends on the underlying molecular structure Φ^* and rotation distribution ρ^* , we sometimes write $\widetilde{m}_k[\Phi^*, \rho^*]$ to highlight its dependence on Φ^* and ρ^* . This notation helps clarify later sections where we consider multiple such distributions.

The method-of-moments approach attempts to reconstruct the molecular structure by matching the empirical moments, computed from the data, to the population moments evaluated at candidate parameters Φ and ρ . These population moments, denoted by $m_k[\Phi, \rho] : \mathbb{R}^{2k} \rightarrow \mathbb{C}$, are defined as

$$m_k[\Phi, \rho](\omega_1, \dots, \omega_k) := \mathbb{E}_\varepsilon \left[\int_{\text{SO}(3)} \widehat{I}_R(\omega_1) \cdots \widehat{I}_R(\omega_{k-1}) \overline{\widehat{I}_R(\omega_k)} \rho(R) dR \right] - B_k(\omega_1, \dots, \omega_k, \sigma), \quad (1.6)$$

where the expectation is taken over the noise term. The reconstruction problem then becomes

$$\min_{\Phi, \rho} \sum_{k=1}^d \lambda_k \|\widetilde{m}_k - m_k[\Phi, \rho]\|_2^2,$$

for some suitably chosen weights $\lambda_k \in \mathbb{R}_{\geq 0}$, where $k = 1, \dots, d$. We often omit the dependence on Φ and ρ from the notation when it is clear from context. In particular, the first and second-order population moments take the following form:

$$m_1(\omega_1) = \mathbb{E}_\varepsilon \left[\int_{\text{SO}(3)} \widehat{I}_R(\omega_1) \rho(R) dR \right], \quad (1.7)$$

$$m_2(\omega_1, \omega_2) = \mathbb{E}_\varepsilon \left[\int_{\text{SO}(3)} \widehat{I}_R(\omega_1) \overline{\widehat{I}_R(\omega_2)} \rho(R) dR \right] - \mathbb{E}_\varepsilon [\widehat{\varepsilon}(\omega_1) \overline{\widehat{\varepsilon}(\omega_2)}], \quad (1.8)$$

where the first expression follows from the assumption that the noise is zero-mean, and the second contains a debias term that depends only on the second-order statistics of the noise. These two moments are the primary focus of this work, as they form the basis for both the theoretical analysis and the proposed reconstruction algorithm.

The method of moments offers several advantages over competing approaches, such as maximum likelihood-based methods with 3-D iterative refinement. One key benefit is that empirical moments in (1.5) can be computed using only one or two passes over the data, after which the raw dataset no longer needs to be accessed [62, 9, 50, 39]. This property significantly reduces computational costs for sufficiently large datasets, especially when compared to the repeated access and high iteration count required in likelihood-based refinement pipelines. Another advantage lies in its applicability to small molecular structures (e.g., below 40 kDa), where state-of-the-art software implementations of 3-D iterative refinement encounter challenges [46, 43]. Even in these challenging regimes, particle locations can still

be detected reliably from micrographs [58, Figure 10f–h], which allows empirical moments to be formed and the method of moments to be applied—bypassing the limitations of traditional refinement.

Historically, the method of moments was introduced to the cryo-EM setting in 1980 by Kam [30], who observed numerically that third-order population moments can uniquely determine bandlimited molecular structures (to be precisely defined in Section 2.1)—under the assumption that the viewing directions are uniformly distributed over $\text{SO}(3)$. These observations have since been rigorously justified under various technical assumptions [5, 22, 21]. Despite these strengths, significant challenges remain. Accurate estimation of the d th-order moment requires a number of samples that scales as $\omega(\sigma^{2d})$ [5, 1], which becomes prohibitive even at moderate moment orders such as $d = 3$, particularly in high-noise settings. Furthermore, when moments are discretized as tensors, both storage and computational costs grow exponentially with the moment order, presenting serious practical limitations. Recent approaches attempt to mitigate these issues by exploiting compressed low-rank tensor formats [28].

To address these complexities, recent efforts have focused on the use of only second-order moments. Under suitable conditions, it has been shown that the second-order moment can determine the molecular structure either uniquely [8, 35, 7, 6] or up to a finite list [48]. These insights have enabled practical algorithms for ab initio reconstruction, which can then be refined using iterative methods, as well as the development of pseudo-metrics for comparing cryo-EM images directly, without explicit reconstruction [61]. Further progress has also been made towards developing efficient numerical schemes that utilize side information, such as knowledge of homologous structures or access to a denoised projection image [10, 29].

Nonetheless, obtaining high-resolution reconstructions directly from moments remains a challenging task. A primary obstacle is the non-convex nature of the moment-matching optimization problem, which can lead iterative methods to stagnate or converge to spurious local minima, and therefore not recover the ground truth structure. Moreover, while some guarantees exist for recovery up to a finite (potentially extremely large) list, certifying exact uniqueness of the underlying structure continues to be an open question in many settings.

1.4 Contributions

This paper introduces a new approach that overcomes prior computational bottlenecks in method-of-moments-based cryo-EM reconstruction by leveraging two second-order moments obtained under distinct orientation distributions. The central idea is to exploit recent experimental advances in cryo-EM sample preparation that enable the collection of multiple datasets of the same underlying molecule, each associated with a different distribution over 3-D orientations.

Specifically, we assume access to empirical second-order moments of the form

$$\tilde{m}_2[\Phi^*, \rho_1^*], \quad \text{and} \quad \tilde{m}_2[\Phi^*, \rho_2^*], \quad (1.9)$$

where ρ_1^* and ρ_2^* denote two distinct underlying rotation distributions that generate the

respective datasets. Throughout this paper, we assume that ρ_1^* is uniform over $\text{SO}(3)$ ², and that ρ_2^* is *in-plane uniform*, meaning that the distribution of the resulting projection images is invariant to image-plane rotations; see (2.6). Without loss of generality, this can be achieved by augmenting the dataset with randomly rotated copies of the projection images. Additionally, we assume that ρ_2^* is *invariant to chirality*, meaning that the distribution of the resulting projection images is unchanged under reflection across a fixed line through the origin in the image plane (e.g., the vertical axis). This property can similarly be enforced via data augmentation by including reflected versions of the projection images with respect to that line. In combination with the in-plane uniformity assumption, this chirality invariance with respect to a single axis implies invariance under reflection across any line through the origin. See Section 2.2 for further structural consequences. We aim to relax the uniformity assumption on ρ_1^* in future work to instead only require that both ρ_1^* and ρ_2^* are in-plane uniform.

Our main contributions are threefold:

- We introduce a new statistical reconstruction framework that leverages two second-order moments computed from datasets with distinct orientation distributions, made possible by recent experimental advances that facilitate collecting multiple datasets of the same molecule under different preparation protocols. To our knowledge, this is the first method to systematically exploit such complementary orientation datasets within the moment-based cryo-EM framework.
- We prove that the molecular structure is uniquely identified by the two second-order moments (rather than up to a finite ambiguity), provided that the structure is bandlimited (see Section 2.4 for a precise statement). This result strengthens prior identifiability guarantees that rely on higher-order moments or strong generative assumptions.
- We design a practical reconstruction algorithm based on convex relaxation, and demonstrate numerically that it accurately recovers bandlimited structures from the population moments. In our experiments, the algorithm consistently converges to the ground truth without stagnation or spurious local minima, enabling efficient and reliable reconstruction.

2 Preliminaries

2.1 Basis representation of structure

Since the image formation model (1.1) involves rotations of a 3-D Fourier volume, it is convenient to represent any such Fourier transformed volume $\hat{\Phi}$ as an element of a function space that is closed under rotations. A natural choice, by the Peter–Weyl theorem (see, e.g., [16]), is the spherical harmonic basis. In particular, we assume $\hat{\Phi}$ is square-integrable and

²We use both ρ_1 and ρ_1^* to denote the uniform/Haar distribution over $\text{SO}(3)$. Readers should note that these notations are equivalent and may appear interchangeably.

supported on a ball of radius $r_{\max}$, and represent it in spherical coordinates (r, θ, φ) using a spherical harmonics expansion for each fixed radius (see, e.g., [48]):

$$\hat{\Phi}(r, \theta, \varphi) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} A_{\ell}^m(r) Y_{\ell}^m(\theta, \varphi), \quad r \in [0, r_{\max}], \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi), \quad (2.1)$$

where Y_{ℓ}^m are the complex-valued spherical harmonics [18, Eq. 14.30.1], the positive integer L is a bandlimit parameter, and $A_{\ell}^m(r)$ is a scalar complex-valued function of r , serving as the expansion coefficients.

To motivate the validity of (2.1), note that the Dirichlet Laplacian on the ball of radius $r_{\max}$ has eigenfunctions $\psi_{k\ell m}(r, \theta, \phi) := j_{\ell}(\lambda_{\ell k} \frac{r}{r_{\max}}) Y_{\ell}^m(\theta, \varphi)$, where j_{ℓ} is the ℓ th spherical Bessel function of the first kind and $\lambda_{\ell k}$ is its k th positive root [26, §3.3], with $k \in \mathbb{Z}_{>0}$, $\ell \in \mathbb{Z}_{\geq 0}$ and $m \in \{-\ell, \dots, \ell\}$. The $\psi_{k\ell m}$'s therefore form a complete orthonormal basis for the space of square-integrable functions on the ball. By summing the $\psi_{k\ell m}$'s over the index k , it follows that any square-integrable function supported on the ball of radius $r_{\max}$ can be represented in the form in (2.1) as we let $L \rightarrow \infty$.

Here we assume that $\hat{\Phi}$ is bandlimited, in the sense that $\hat{\Phi}$ can be represented exactly in the form in (2.1) for a finite value of L . Although this is an idealization, should this not hold, higher components $A_{\ell}^m(r) Y_{\ell}^m(\theta, \Phi)$ for $\ell > L$ can be treated as an additional source of approximation error.

Section (2.3) presents explicit analytical expressions for the moment $m_2[\Phi, \rho]$ for bandlimited functions of the form (2.1), using the complex-valued spherical harmonics introduced above. When describing our algorithm, it will however be convenient to also use real-valued spherical harmonics. We therefore introduce notation for converting complex-valued spherical harmonics basis coefficients into coefficients in the real-valued spherical harmonics basis.

For each $\ell \in \mathbb{Z}_{\geq 0}$, write $Q_{\ell} \in \mathbb{C}^{(2\ell+1) \times (2\ell+1)}$ as a unitary matrix that has non-zero entries only on the main diagonal and anti-diagonal, with non-zero components defined explicitly by

$$(Q_{\ell})_{mm} = \begin{cases} i/\sqrt{2}, & \text{if } m < 0, \\ 1 & \text{if } m = 0, \\ (-1)^m/\sqrt{2}, & \text{if } m > 0, \end{cases} \quad (Q_{\ell})_{-m,m} = \begin{cases} 1/\sqrt{2}, & \text{if } m < 0, \\ 1 & \text{if } m = 0, \\ -(-1)^m i/\sqrt{2}, & \text{if } m > 0, \end{cases} \quad (2.2)$$

for all $-\ell \leq m \leq \ell$. We denote by $Y_{\ell m}(\theta, \varphi)$ the real-valued spherical harmonics defined by the convention

$$Y_{\ell m}(\theta, \varphi) = \sum_{m'=-\ell}^{\ell} (Q_{\ell})_{m,m'} Y_{\ell}^{m'}(\theta, \varphi), \quad (2.3)$$

where the real-valuedness of $Y_{\ell m}$ follows from [18, Eq. 14.30.1, Eq. 14.9.3]. By changing the angular basis to $Y_{\ell m}(\theta, \varphi)$, we can equivalently write (2.1) as

$$\hat{\Phi}(r, \theta, \varphi) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} A_{\ell m}(r) Y_{\ell m}(\theta, \varphi), \quad (2.4)$$

where $A_{\ell m}(r)$ is the corresponding expansion coefficient in the real-valued spherical harmonics basis, obtained by linearly transforming the complex coefficients using the unitary matrix Q_ℓ :

$$A_{\ell m}(r) = \sum_{m'=-\ell}^{\ell} A_{\ell}^{m'}(r) (\overline{Q}_\ell)_{mm'}.$$

It is equivalent and more convenient to write in the opposite direction that

$$A_{\ell}^m(r) = \sum_{m'=-\ell}^{\ell} A_{\ell m'}(r) (Q_\ell)_{m'm}, \quad (2.5)$$

which can explicitly be written out as

$$A_{\ell}^m(r) = \begin{cases} \frac{i}{\sqrt{2}} A_{\ell m}(r) + \frac{1}{\sqrt{2}} A_{\ell, -m}(r), & \text{if } m < 0, \\ A_{\ell 0}, & \text{if } m = 0, \\ \frac{-(-1)^m i}{\sqrt{2}} A_{\ell, -m}(r) + \frac{(-1)^m}{\sqrt{2}} A_{\ell m}(r), & \text{if } m > 0. \end{cases}$$

Remark 2.1. For real-valued structures Φ , symmetry properties of the Fourier transform imply that the coefficients $\{A_{\ell m}(r) : -\ell \leq m \leq \ell\}$ are real for even ℓ and purely imaginary for odd ℓ [8].

2.2 Basis representation of rotation distribution

We model the distribution of viewing orientations as having a density function $\rho(R)$ with respect to the uniform/Haar measure over the rotation group $\text{SO}(3)$. To facilitate analysis and to reflect symmetries commonly present—or enforceable via data augmentation—in cryo-EM experiments, we work under the following two structural assumptions on $\rho(R)$:

1. **In-plane uniformity:** ρ is invariant under right multiplication by any rotation about the z -axis, i.e.,

$$\rho(R) = \rho(Rz(\alpha)), \quad (2.6)$$

for all $R \in \text{SO}(3)$ and rotations $z(\alpha)$ of $\alpha \in \mathbb{R}$ radians around the z -axis. This implies that the distribution of projection images is invariant to image-plane 2-D rotations.

2. **Chirality invariance:** ρ is invariant under conjugation by the reflection matrix $J = \text{diag}(1, 1, -1)$, i.e.,

$$\rho(R) = \rho(JRJ), \quad (2.7)$$

for all $R \in \text{SO}(3)$. This implies that the projection image distribution is invariant under reflection across a fixed axis (e.g., the vertical axis) in the image plane; combined with in-plane uniformity, this further implies invariance under reflection across *any* line through the origin.

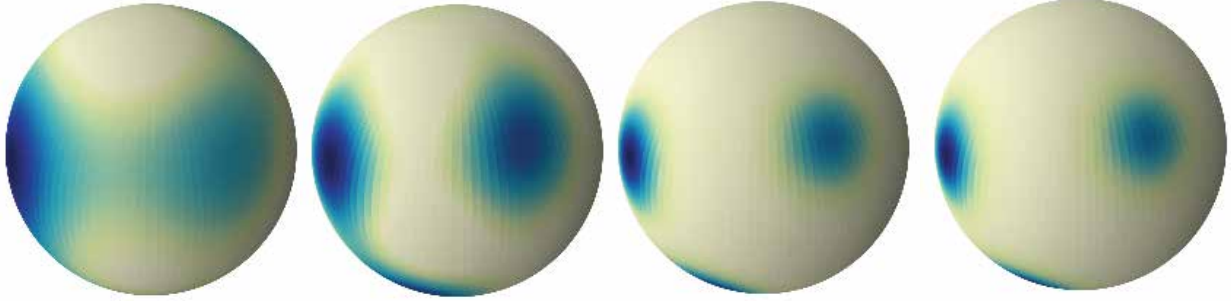


Figure 2.1: Illustration of the basis expansion of (2.8) as function of $\theta(R)$ and $\varphi(R)$, using $P = 3, 5, 10$ from leftmost to second-most right figure, and the rightmost figure as the ground-truth.

We show in Appendix A.1 that any square-integrable density $\rho(R)$ on $\text{SO}(3)$ inducing an in-plane uniform distribution of viewing orientations (of the tomographic projection images) can be effectively represented using a bandlimited expansion of order P :

$$\rho(R)dR = \sum_{p=0}^P \sum_{u=-p}^p B_{p,u} U_{u0}^p(R) dR, \quad (2.8)$$

where dR is the uniform/Haar measure on $\text{SO}(3)$, and $U_{u0}^p(R)$ are Wigner U -matrix entries given explicitly by [16, Eq. 9.42]

$$U_{u0}^p(R) = (-1)^u \sqrt{\frac{4\pi}{2p+1}} \overline{Y_p^u(\theta(R), \varphi(R))}, \quad (2.9)$$

with $(\theta(R), \varphi(R))$ the spherical angles of the third column of R . The cutoff P is the bandlimit of the distribution ρ ; its relationship with the volume bandlimit L will be discussed further in Section 2.4 and Section 4.

The complex-valued basis coefficients $B_{p,u}$ satisfy constraints from the real-valued nature and normalization of ρ :

$$\overline{B_{p,u}} = (-1)^u B_{p,-u}, \quad \text{and} \quad B_{0,0} = 1, \quad (2.10)$$

and chirality invariance further implies

$$B_{p,u} = 0, \quad \text{for all odd } p. \quad (2.11)$$

These conditions yield a finite-dimensional representation of ρ whose coefficients lie in a structured subset of complex-valued vectors.

2.3 Analytic expressions for moments

We show in Appendix A.2.2 that the second-order population moment (1.6) for a bandlimited structure of the form (2.1) with a rotation distribution of the form (2.8), has a succinct form.

To present it, denote by $A_\ell : [0, r_{\max}] \rightarrow \mathbb{C}^{1 \times (2\ell+1)}$, for $\ell \in \{0, \dots, L\}$, $n \in \{-L, \dots, L\}$ the row-vector-valued function defined by

$$A_\ell(r) = [A_\ell^{-\ell}(r), A_\ell^{-\ell+1}(r), \dots, A_\ell^\ell(r)], \quad (2.12)$$

and define the matrix $\mathcal{B}_{\ell, \ell'}^n \in \mathbb{C}^{(2\ell+1) \times (2\ell'+1)}$, for $\ell, \ell' \in \{0, \dots, L\}$, $n \in \{-L, \dots, L\}$, by

$$(\mathcal{B}_{\ell, \ell'}^n)_{m, m'} = (-1)^{m+n} \mathcal{N}_\ell^n \mathcal{N}_{\ell'}^n \sum_{\ell''=\max\{|m-m'|, |\ell-\ell'|\}}^{\min\{\ell+\ell', P\}} \frac{\mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n)}{2\ell''+1} B_{\ell'', m'-m}, \quad (2.13)$$

for $-\ell \leq m \leq \ell$ and $-\ell' \leq m' \leq \ell'$. Here, the scalar $\mathcal{N}_\ell^n \in \mathbb{R}$ is defined by

$$\mathcal{N}_\ell^n = N_\ell^n \cdot \mathbf{1}_{\{\ell \equiv n \pmod{2}\}} \cdot \mathbf{1}_{\{\ell \geq |n|\}}, \quad (2.14)$$

where

$$N_\ell^n = \sqrt{\frac{2\ell+1}{4\pi}} \sqrt{\frac{(\ell-n)!}{(\ell+n)!}} \cdot P_\ell^n(0), \quad (2.15)$$

with $P_\ell^n(x)$ denoting the associated Legendre functions, [18, Eq. (14.3.6)]. The $\mathcal{C}_{\ell''}$ are septuply indexed constants defined (for general indices) by

$$\mathcal{C}_{\ell''}(\ell, \ell', m, m', n, n') := C(\ell, m; \ell', m' | \ell'', m+m') C(\ell, n; \ell', n' | \ell'', n+n'), \quad (2.16)$$

where the $C(\cdot, \cdot; \cdot, \cdot | \cdot, \cdot)$ are Clebsch-Gordan coefficients [16, Section 9.9].

With these definitions in place, we are now ready to state the following proposition giving the expression for the second-order moment. The proof is deferred to Appendix A.2.2.

Proposition 2.2. *The second-order population moment (1.6) for a bandlimited structure of the form (2.1) with a rotation distribution of the form (2.8) can be written as*

$$m_2(r, \varphi, r', \varphi') = \sum_{n=-L}^L e^{in(\varphi-\varphi')} \sum_{\ell=0}^L \sum_{\ell'=0}^L A_\ell(r) \mathcal{B}_{\ell, \ell'}^n (A_{\ell'}(r'))^H. \quad (2.17)$$

In particular, $\mathcal{B}_{\ell, \ell'}^n$ satisfies the Hermitian property $\mathcal{B}_{\ell, \ell'}^n = (\mathcal{B}_{\ell', \ell}^n)^H$. Also, $\mathcal{B}_{\ell, \ell'}^n = 0$ for any $-L \leq n \leq L$ whenever $\ell \not\equiv \ell' \pmod{2}$.

Remark 2.3. From the expression of $\mathcal{B}_{\ell, \ell'}^n$ in (2.13) and the expansion of m_2 in (2.17), we observe that the coefficients $B_{p,u}$ with $p > 2L$ do not appear in m_2 , due to the constraint $\ell'' \leq \min\{\ell + \ell', P\}$ with $\ell, \ell' \leq L$. Similarly, a direct inspection of the first moment m_1 (see Appendix A.2.1) reveals that it depends only on $B_{p,u}$ with $p \leq L$. Therefore, the coefficients $B_{p,u}$ for all $p > 2L$ do not influence the population statistics considered in this work.

Remark 2.4. In particular, for the uniform distribution of rotations, $B_{\ell,m} = 0$ unless $\ell = 0$ and $m = 0$. Applying (2.15), the parity property of P_ℓ^n , and the explicit formula for Clebsch-Gordan coefficients

$$C(\ell, m; \ell, -m|0, 0) = \frac{(-1)^{\ell-m}}{\sqrt{2\ell+1}}, \quad C(\ell, n; \ell, -n|0, 0) = \frac{(-1)^{\ell-n}}{\sqrt{2\ell+1}},$$

then (2.17) simplifies to

$$m_2(r, r', \psi) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} A_\ell^m(r) \overline{A_\ell^m(r')} \sum_{n=-\ell}^{\ell} \frac{1}{4\pi} \frac{(\ell-n)!}{(\ell+n)!} (P_\ell^n(0))^2 e^{in\psi}.$$

Further using the addition theorem for spherical harmonics [16, Eqn. 4.37], we reach

$$m_2(r, r', \psi) = \frac{1}{4\pi} \cdot \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} A_\ell^m(r) \overline{A_\ell^m(r')} P_\ell(\cos \psi), \quad (2.18)$$

where the P_ℓ 's are the Legendre polynomials [18, §18.3], and $\psi := \varphi - \varphi'$.

2.4 Formal problem statement

We formally formulate the problem of recovering the ground-truth structure Φ^* from its associated moment observations. We make the following assumptions.

Assumption 2.5. (Assumptions on the structure Φ and the distributions ρ_1, ρ_2)

1. Φ is bandlimited with respect to its angular variables, i.e., Φ can be expressed as in (2.1) for some $L \geq 3$.
2. The radial functions $(A_\ell^m(r) : 0 \leq \ell \leq L, -\ell \leq m \leq \ell)$ in (2.1) are linearly independent.
3. The distribution ρ_1 for the first dataset is the uniform/Haar measure on $\text{SO}(3)$.
4. The distribution ρ_2 for the second dataset is in-plane uniform and invariant to chirality (see Section 2.2 for precise definitions). Moreover, ρ_2 is bandlimited with cutoff $P \geq 2L$, and its expansion coefficients from (2.8)

$$(B_{p,u} : 1 \leq p \leq 2L, \ p \text{ even}, \ -p \leq u \leq p) \in \mathbb{C}^{2L^2+3L}$$

are Zariski-generic.

The 3-D reconstruction problem can be formally stated as follows.

Problem 2.6. Let $\Phi^*, \rho_1^*, \rho_2^*$ satisfy Assumption 2.5. Given samples of $\tilde{m}_1[\Phi^*, \rho_1^*]$, $\tilde{m}_2[\Phi^*, \rho_1^*]$ and $\tilde{m}_2[\Phi^*, \rho_2^*]$ on a grid in polar coordinates defined by

$$\{r_1, \dots, r_{M_r}\} \subseteq [0, r_{\max}], \quad \{\varphi_1, \dots, \varphi_{M_\varphi}\} \subseteq [0, 2\pi), \quad (2.19)$$

recover the structure Φ^* .

Remark 2.7. In cryo-EM, datasets with non-uniform viewing directions are readily obtained. Experimental or data processing techniques are generally required to obtain uniform distributions. In techniques like XFEL, distributions close to uniform are more easily available [45, 32, 19, 33].

3 Algorithm: Method of Double Moments

We present the main computational result: an algorithm for solving Problem 2.6 given access only to sample moments evaluated on a discrete polar grid. The algorithm consists of three steps:

1. **Kam’s method (cf. [30]):** Use the uniform sample moment $\tilde{m}_2[\Phi^*, \rho_1^*]$ to recover the structure coefficients $A_\ell(r_i)$ for $i = 1, \dots, M_r$, up to the action of unknown orthogonal matrices $O_\ell \in O(2\ell + 1)$ for $\ell = 0, \dots, L$, i.e., the problem reduces to estimating the set $\{O_\ell\}$.
2. **Formulate optimization:** Use the non-uniform sample moment $\tilde{m}_2[\Phi^*, \rho_2^*]$ to set up a non-linear least-squares problem in the variables $\{O_\ell\}$ and $\{B_{p,u}\}$.
3. **Convex relaxation and alternating refinement:** Solve the optimization problem by iteratively updating $\{O_\ell\}$ and $\{B_{p,u}\}$, while regularizing the constraint using information from the first-order moment $\tilde{m}_1[\Phi^*, \rho_1^*]$.

These steps are detailed in Sections 3.1 – 3.3, respectively. Algorithm 1 presents the complete procedure in summary form. Here, all quantities with a widetilde $\tilde{\cdot}$ are empirically accessible from the observed samples.

3.1 Step 1: Kam’s method via uniform sample moment

Kam [30] showed, via the Fourier slice theorem, that if the viewing directions are uniformly distributed over $SO(3)$, then the autocorrelation function of the 3-D volume with itself can be recovered directly from the second-order moment of 2-D projection images. In particular, for a uniform distribution ρ_1^* , it follows from (2.18) and the orthogonality of the Legendre polynomials that, for each $\ell = 0, \dots, L$ and $r, r' \in [0, r_{\max}]$,

$$\begin{aligned} C_\ell(r, r') &:= 2\pi(2\ell + 1) \cdot \int_0^\pi m_2[\Phi, \rho_1^*](r, r', \psi) P_\ell(\cos \psi) \sin(\psi) d\psi \\ &= \sum_{m=-\ell}^{\ell} A_\ell^m(r) \overline{A_\ell^m(r')} = \sum_{m=-\ell}^{\ell} A_{\ell m}(r) A_{\ell m}(r'), \end{aligned} \quad (3.1)$$

where the last equation comes from the unitarity of Q_ℓ in Section 2.1. This relation provides partial information about the unknown coefficients A_ℓ^n or $A_{\ell m}$.

Assume we have access to samples on the grid in (2.19). We approximate the integral in (3.1) by first using the sample moment $\tilde{m}_2[\Phi^*, \rho_1^*]$ instead of the population moment, and

Algorithm 1: Method of Double Moments

Input: Sample moments $\tilde{m}_1[\Phi^*, \rho_1^*]$, $\tilde{m}_2[\Phi^*, \rho_1^*]$, and $\tilde{m}_2[\Phi^*, \rho_2^*]$ satisfying Assumption 2.5 and sampled on the grid (2.19), maximum number of iterations K .

Result: Approximation $\Phi^{(K)}$ to the underlying molecular structure Φ^* .

- 1 Compute $\tilde{C}$ in (3.2) through numerical quadrature, and obtain $\tilde{A}_\ell$ from $\tilde{C}_\ell$ via Cholesky decomposition followed by taking the real part. /* Step 1 */
- 2 Compute $\tilde{G}^n$ in (3.8) through numerical quadrature, and obtain $\tilde{\mathcal{M}}^n$ in (3.11) by left- and right-multiplying $\tilde{G}^n$ by $\tilde{A}^\dagger$ and $\tilde{A}^{\text{H},\dagger}$, respectively. /* Step 2 */
- 3 Compute iterative solutions $O_\ell^{(k)}$ and $B_{p,u}^{(k)}$ to the update equations (3.19) – (3.21), for $k = 1, \dots, K$. /* Step 3 */
- 4 Return the inverse discrete Fourier transform of

$$\hat{\Phi}^{(K)}(r_i, \theta, \varphi) := \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} (\tilde{A}_\ell O_\ell^{(K)})_{i,m} Y_{\ell m}(\theta, \varphi),$$

sampled at the radial points r_i from (2.19) for $i = 1, \dots, M_r$.

second by using numerical quadrature with quadrature points φ_s, φ_q and quadrature weights $w_{s,q}$, for $s, q = 0, \dots, M_\varphi$, i.e., we compute

$$\tilde{C}_\ell(r_i, r_j) := 2\pi(2\ell + 1) \cdot \sum_{s,q=0}^{M_\varphi} w_{s,q} \tilde{m}_2[\Phi^*, \rho_1^*](r_i, r_j, \varphi_s - \varphi_q) P_\ell(\cos(\varphi_s - \varphi_q)) \sin(\varphi_s - \varphi_q). \quad (3.2)$$

Next, for each $\ell = 0, \dots, L$, we introduce the matrices $A_\ell \in \mathbb{C}^{M_r \times (2\ell+1)}$, $\check{A}_\ell \in \mathbb{C}^{M_r \times (2\ell+1)}$, $C_\ell \in \mathbb{R}^{M_r \times M_r}$, and $\tilde{C}_\ell \in \mathbb{C}^{M_r \times M_r}$. They are defined entrywise as

$$(A_\ell)_{k,m} = A_\ell^m(r_k), \quad (\check{A}_\ell)_{k,m} = A_{\ell m}(r_k), \quad (C_\ell)_{i,j} = C_\ell(r_i, r_j), \quad (\tilde{C}_\ell)_{i,j} = \tilde{C}_\ell(r_i, r_j), \quad (3.3)$$

respectively, where $\{r_1, \dots, r_{M_r}\}$ denote the sampling radial points. Inserting these definitions into (3.1) and (3.2) gives

$$\tilde{C}_\ell \approx C_\ell = A_\ell A_\ell^{\text{H}} = \check{A}_\ell \check{A}_\ell^{\text{H}}, \quad (3.4)$$

where the approximation reflects discretization and sampling errors.

Recall from Remark 2.1 that $\check{A}_\ell$ is real for even ℓ and purely imaginary for odd ℓ . Motivated by this structure, applying the Cholesky decomposition of the matrix C_ℓ , we obtain factors

$$\check{A}_\ell \in \mathbb{C}^{M_r \times (2\ell+1)},$$

which can be chosen real for even ℓ and purely imaginary for odd ℓ . Moreover, there exists a unique real orthogonal matrix $O_\ell \in \mathbf{O}(2\ell + 1)$ such that

$$\check{A}_\ell = \check{A}_\ell O_\ell^\top = A_\ell Q_\ell^\mathbf{H} O_\ell^\top. \quad (3.5)$$

In practice, we replace C_ℓ with its sample counterpart $\tilde{C}_\ell$, perform a Cholesky decomposition, and take the real part to obtain an estimate $\tilde{A}_\ell$ for A_ℓ . The remaining task is to recover the orthogonal matrices $\{O_\ell\}$, which would in turn allow us to fully estimate $\{A_\ell\}$.

3.2 Step 2: Formulate optimization with the non-uniform sample moment

The remainder of the computational method attempts to recover the matrices O_ℓ , for $0 \leq \ell \leq L$ by matching the additional non-uniform sample moment $\tilde{m}_2[\Phi^*, \rho_2^*]$ to the corresponding population moment. We therefore re-express $m_2[\Phi, \rho_2]$ in terms of the matrices $\check{A}_\ell$ computed in Section 3.1. Inserting (3.5) into (2.17) gives

$$m_2(r, \varphi, r', \varphi') = \sum_{n=-L}^L e^{in(\varphi-\varphi')} \sum_{\ell=0}^L \sum_{\ell'=0}^L \check{A}_\ell(r) O_\ell Q_\ell \mathcal{B}_{\ell,\ell'}^n Q_{\ell'}^\mathbf{H} O_{\ell'}^\top (\check{A}_{\ell'}(r'))^\mathbf{H}, \quad (3.6)$$

where we denote by $\check{A}_\ell(r)$ the row of $\check{A}_\ell$ corresponding to the radial sample r .

Note that performing the integration over φ and φ' yields

$$\frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} m_2[\Phi, \rho_2](r, \varphi, r', \varphi') e^{-in(\varphi-\varphi')} d\varphi d\varphi' = \sum_{\ell=0}^L \sum_{\ell'=0}^L \check{A}_\ell(r) O_\ell Q_\ell \mathcal{B}_{\ell,\ell'}^n Q_{\ell'}^\mathbf{H} O_{\ell'}^\top (\check{A}_{\ell'}(r'))^\mathbf{H}.$$

Thus, the integral can be approximated numerically by first replacing m_2 with the sample moment $\tilde{m}_2[\Phi^*, \rho_2^*]$ and then applying numerical quadrature, which yields

$$\begin{aligned} \tilde{G}^n(r_i, r_j) &:= \frac{1}{(2\pi)^2} \sum_{s,q=0}^{M_\varphi} w_{s,q} \tilde{m}_2[\Phi^*, \rho_2^*](r_i, r_j, \varphi_s - \varphi_q) e^{-in(\varphi_s - \varphi_q)} \\ &\approx \sum_{\ell,\ell'=0}^L \check{A}_\ell(r_i) O_\ell Q_\ell \mathcal{B}_{\ell,\ell'}^n Q_{\ell'}^\mathbf{H} O_{\ell'}^\top (\check{A}_{\ell'}(r_j))^\mathbf{H}, \end{aligned} \quad (3.7)$$

for each $-L \leq n \leq L$, where the approximate equality arises from discretization and sampling errors.

Define the matrices $\tilde{G}^n \in \mathbb{C}^{M_r \times M_r}$ and $\mathcal{M}_{\ell,\ell'}^n \in \mathbb{C}^{(2\ell+1) \times (2\ell'+1)}$ by

$$(\tilde{G}^n)_{i,j} = \tilde{G}^n(r_i, r_j), \quad \mathcal{M}_{\ell,\ell'}^n = O_\ell Q_\ell \mathcal{B}_{\ell,\ell'}^n Q_{\ell'}^\mathbf{H} O_{\ell'}^\top. \quad (3.8)$$

Let $\mathcal{M}^n$ and $\mathcal{B}^n \in \mathbb{C}^{(L+1)^2 \times (L+1)^2}$ be the block matrices whose (ℓ, ℓ') -th blocks are $\mathcal{M}_{\ell, \ell'}^n$ and $\mathcal{B}_{\ell, \ell'}^n$, respectively, i.e.,

$$\mathcal{M}^n = \begin{bmatrix} \mathcal{M}_{0,0}^n & \cdots & \mathcal{M}_{0,L}^n \\ \vdots & \ddots & \vdots \\ \mathcal{M}_{L,0}^n & \cdots & \mathcal{M}_{L,L}^n \end{bmatrix}, \quad \mathcal{B}^n = \begin{bmatrix} \mathcal{B}_{0,0}^n & \cdots & \mathcal{B}_{0,L}^n \\ \vdots & \ddots & \vdots \\ \mathcal{B}_{L,0}^n & \cdots & \mathcal{B}_{L,L}^n \end{bmatrix}. \quad (3.9)$$

Further define the block matrices

$$\check{\mathcal{A}} = [\check{A}_0 \quad \cdots \quad \check{A}_L], \quad \tilde{\mathcal{A}} = [\tilde{A}_0 \quad \cdots \quad \tilde{A}_L] \in \mathbb{C}^{M_r \times (L+1)^2},$$

where $\check{A}_\ell$ are the ideal factors and $\tilde{A}_\ell$ their estimates obtained from $\tilde{C}_\ell$. Then the approximate relation in (3.7) can be written as

$$\tilde{G}^n \approx \check{\mathcal{A}} \mathcal{M}^n \check{\mathcal{A}}^H \approx \tilde{\mathcal{A}} \mathcal{M}^n \tilde{\mathcal{A}}^H. \quad (3.10)$$

In the expression above, the sample moment information provides us with $\tilde{G}^n$ and $\tilde{\mathcal{A}}$, whereas our goal is to recover the unknown matrices $\{O_\ell\}$ and $\{\mathcal{B}_{\ell, \ell'}^n\}$ contained within $\mathcal{M}^n$.

Assuming $\tilde{\mathcal{A}}$ has full rank (which in particular requires that $M_r \geq (L+1)^2$), we can left- and right-multiply this equation by the pseudoinverse matrices $\tilde{\mathcal{A}}^\dagger$ and $\tilde{\mathcal{A}}^{H, \dagger}$ to obtain approximate access to $\mathcal{M}^n$. Writing

$$\mathcal{O} = \text{blockdiag}_{\ell=0, \dots, L}(O_\ell), \quad \text{and} \quad \mathcal{Q} = \text{blockdiag}_{\ell=0, \dots, L}(Q_\ell),$$

we obtain

$$\tilde{\mathcal{A}}^\dagger \tilde{G}^n \tilde{\mathcal{A}}^{H, \dagger} := \tilde{\mathcal{M}}^n \approx \mathcal{M}^n = \mathcal{O} \mathcal{Q} \mathcal{B}^n \mathcal{Q}^H \mathcal{O}^T, \quad \text{for} \quad -L \leq n \leq L. \quad (3.11)$$

This naturally leads to a least-squares problem, where one seeks to find the matrices $\{O_\ell\}$ and parameters $\{B_{p,u}\}$ by minimizing

$$\sum_{n=-L}^L \left\| \tilde{\mathcal{M}}^n - \mathcal{O} \mathcal{Q} \mathcal{B}^n \mathcal{Q}^H \mathcal{O}^T \right\|^2. \quad (3.12)$$

3.3 Step 3: Convex relaxation and alternating refinement

We solve the problem (3.12) through an alternating procedure for the matrices O_ℓ and the parameters $B_{p,u}$, by initializing $O_\ell^{(0)}$ as arbitrary orthogonal matrices and updating the parameters $\{B_{p,u}^{(k)}\}$ and $\{O_\ell^{(k)}\}$ iteratively, for $k = 0, \dots, K$.

Solving for $\{B_{p,u}\}$. Given $\mathcal{O}^{(k)} = \text{blockdiag}_{\ell=0,\dots,L}(O_\ell^{(k)})$, the update of $\{B_{p,u}\}$ proceeds by minimizing the expression

$$\sum_{n=-L}^L \left\| \tilde{\mathcal{M}}^n - \mathcal{O}^{(k)} \mathcal{Q} \mathcal{B}^n \mathcal{Q}^H \mathcal{O}^{(k),\top} \right\|^2, \quad (3.13)$$

in the variables $\{B_{p,u}\}$ while imposing the constraints (2.10), i.e., by restricting B to the set

$$\mathcal{S}_B := \left\{ \{B_{p,u}\}_{(p,u) \in \mathcal{I}} : B_{p,u} \in \mathbb{C}, \quad \overline{B_{p,u}} = (-1)^u B_{p,-u}, \quad B_{0,0} = 1 \right\}, \quad (3.14)$$

where $\mathcal{I} := \{(p,u) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z} : -p \leq u \leq p, p \leq P\}$. Note that the constraint set $\mathcal{S}_B$ does not impose positivity of the resulting density in (2.8), for simplicity. This can however be incorporated by, for instance, requiring positivity at a given set of collocation points R_i (see [48, Eq. (51)]). This yields the constraints $\rho(R_i) \geq 0$, which are linear constraints in the variables $\{B_{p,u}\}$ and therefore can be included when minimizing (3.13), although at the expense of increased runtime.

Solving for $\{O_\ell\}$. Given the coefficients $B_{p,u}^{(k)}$, we obtain $\mathcal{B}_{\ell,\ell'}^{n,(k)}$ from (2.13) and form the block matrix $\mathcal{B}^{n,(k)}$ as in (3.9). The update of $\{O_\ell\}$ then follows a relaxed procedure.

First, note that

$$\left\| \tilde{\mathcal{M}}^n - \mathcal{O} \mathcal{Q} \mathcal{B}^{n,(k)} \mathcal{Q}^H \mathcal{O}^\top \right\| = \left\| \tilde{\mathcal{M}}^n \mathcal{O} - \mathcal{O} \mathcal{Q} \mathcal{B}^{n,(k)} \mathcal{Q}^H \right\|. \quad (3.15)$$

By relaxing the orthogonality constraint on $\mathcal{O}$, we introduce an unconstrained surrogate

$$\mathcal{X} = \text{blockdiag}_{\ell=0,\dots,L}(X_\ell)$$

for $\mathcal{O}$, where each X_ℓ has the same matrix size as O_ℓ . We then solve for $\mathcal{X}$ in a least-squares sense and subsequently orthogonalize the result to recover $\mathcal{O}$. Specifically, we restrict the search to the following set:

$$\mathcal{S}_\mathcal{X} = \left\{ \{X_\ell\}_{\ell=0,\dots,L} : X_\ell \in \mathbb{R}^{(2\ell+1) \times (2\ell+1)}, X_0 = 1, X_1 = I_3 \right\}. \quad (3.16)$$

This restriction is without loss of generality, because for any choice of X_0 and X_1 , there exists a rotation of the underlying structure Φ^* that maps X_0 to 1 and X_1 to I_3 simultaneously (see [16, pg. 324]).

Given a solution $\mathcal{X}$ in $\mathcal{S}_\mathcal{X}$, we orthogonalize it by projecting onto

$$\mathcal{S}_\mathcal{O} = \left\{ \{O_\ell\}_{\ell=0,\dots,L} : O_\ell \in \mathbb{O}(2\ell+1) \right\}, \quad (3.17)$$

which is equivalent to solving the constrained least-squares problem

$$\min_{\{O_\ell\} \in \mathcal{S}_\mathcal{O}} \|\mathcal{O} - \mathcal{X}\|^2 = \min_{\{O_\ell\} \in \mathcal{S}_\mathcal{O}} \sum_{\ell=0}^L \|O_\ell - X_\ell\|^2, \quad (3.18)$$

where the equality follows from the block-diagonal structures of $\mathcal{X}$ and $\mathcal{O}$. Each term in the sum can be solved efficiently using the orthogonal Procrustes procedure, which admits a closed-form solution by computing the singular value decomposition (SVD) of each diagonal block X_ℓ [47].

Summary. Taken together, the update equations for $\{B_{p,u}\}$ and $\{O_\ell\}$ can be written as

$$\{B_{p,u}^{(k+1)}\}_{(p,u) \in \mathcal{I}} = \arg \min_{\{B_{p,u}\} \in \mathcal{S}_{\mathcal{B}}} \sum_{n=-L}^L \|\widetilde{\mathcal{M}}^n - \mathcal{O}^{(k)} \mathcal{Q} \mathcal{B}^n \mathcal{Q}^H \mathcal{O}^{(k),\top}\|^2, \quad (3.19)$$

$$\{X_\ell^{(k+1)}\}_{\ell=0,\dots,L} = \arg \min_{\{X_\ell\} \in \mathcal{S}_{\mathcal{X}}} \sum_{n=-L}^L \|\widetilde{\mathcal{M}}^n \mathcal{X} - \mathcal{X} \mathcal{Q} \mathcal{B}^{n,(k)} \mathcal{Q}^H\|^2, \quad (3.20)$$

$$\{O_\ell^{(k+1)}\}_{\ell=0,\dots,L} = \arg \min_{\{O_\ell\} \in \mathcal{S}_{\mathcal{O}}} \sum_{\ell=0}^L \|O_\ell - X_\ell^{(k+1)}\|^2. \quad (3.21)$$

Equations (3.19) and (3.20) are structured least squares-problems in the variables $B_{p,u}$ and X_ℓ , respectively, and can be solved efficiently. The orthogonalization procedure in equation (3.21) is a sequence of $L + 1$ orthogonal Procrustes problems.

3.4 Computational complexity

Estimation of the second order moments from N images of size $M \times M$ takes $O(NM^3 + M^4)$ operations [39]. Step 1 of Algorithm 1 takes $O(LM_r^2M_\varphi^2)$ operations and step 2 has complexity $O(M_r^2L^3)$. Step 3 takes $O(LM_r^2M_\varphi^2)$ operations. Forming the matrices $\mathcal{B}^n$ and $\check{\mathcal{A}}$ takes time $O(L^2)$ and $O(M_rL^2)$, respectively. Forming the matrices $\mathcal{O}^{(k)} \mathcal{Q} \mathcal{B}^n \mathcal{Q}^H \mathcal{O}^{(k),\top}$ and $\mathcal{O} \mathcal{Q} \mathcal{B}^{n,(k)} \mathcal{Q}^H \mathcal{O}^\top$ in each iteration takes time $O(L \sum_{\ell,\ell'=0}^L \ell^2 \ell' + \ell \ell'^2) = O(L^6)$, for a total of $O(KL^6)$ operations when running K iterations. Solving the update equation for the parameters $\{B_{p,u}\}$ takes time $O(L^6 + L^5P^4 + P^3) = O(L^9)$, when using all $O(L^5)$ available equations for the $O(L^2)$ variables. Solving the update equation for X takes time $O(L^6 + L^5L^6 + L^9) = O(L^{11})$, when using all $O(L^5)$ available equations for the $O(L^3)$ variables. Solving the update equation for the O_ℓ takes time $O(\sum_{\ell=0}^L \ell^3) = O(L^4)$.

The above complexities do not take several structural features into account. For instance, it is possible to use fewer than the $O(L^5)$ equations to solve for the $O(L^3)$ variables when updating X . Moreover, the linear systems exhibit sparsity and iterative solvers can therefore be used for lower complexity (and this is an option in the accompanying code). Lastly, the linear systems have additional structure and are in fact coupled Sylvester-type equation for which there are specialized linear solvers with improved complexity.

By comparison, the complexity of one iteration in the EM algorithm is $O(NM^4 \log M)$ (ignoring translational shifts). For $L = O(1)$, the method-of-double-moments therefore costs less than EM by a factor of $O(N \log M)$ per iteration. Moreover, the upfront cost of moment formation is less than the complexity of one EM iteration by a factor of $O(M \log M)$.

3.5 Numerical experiments

This section demonstrates the Algorithm of Section 3 using the two simulated datasets EMD-2660 [59] and EMD-32743 [27] from the online electron microscopy data bank [34].

For varying M , we generate 120000 projection images of size $M \times M$, with signal-to-noise-ratio 0.1 and affected by convolution with point spread functions with 10000 defocus groups. The projection images were generated with voltage 300 kV, defocus values ranging between 1 μm and 3 μm and spherical aberration 2 mm. The ground truth distribution of viewing angles was taken to be a mixture of 8 Langevin distributions.

The result of running Alg. 1 on bandlimited structures is shown in Figures 3.1 and 3.2. Notably, the algorithm successfully recovers bandlimited functions up to a maximum bandlimit $L(M)$ that depends on the number of pixels in the projection images. This stems from the inversion of the linear system in Eq. (3.11), which is only well-posed for a limited range of bandlimits L , for a fixed value of M .

4 Uniqueness theorem

We give the main theoretical result of this paper. Theorem 4.1 ensures that the first and second population moments of the 2-D images *generically uniquely* identify the structure Φ and the non-uniform distribution ρ_2 (i.e., the fourth item of Assumption 2.5), when ρ_2 is suitably low-passed. Here, *generic* refers to a generic condition on the distribution ρ_2 , and *unique* emphasizes the unique recovery of the structure Φ (in contrast to list-recovery formulations; see, e.g., [48, 5]). This theorem provides a fundamental guarantee for the computational problem stated in Problem 2.6. In the statement below, we denote by $\rho_2^{\downarrow 2L}$ the low-pass of ρ_2 to degree $2L$, i.e., (2.8) truncated to $p \leq 2L$.

Theorem 4.1. *Assume Φ , ρ_1 , and ρ_2 satisfy Assumption 2.5. Then population moments $m_1[\Phi, \rho_1]$, $m_2[\Phi, \rho_1]$ and $m_2[\Phi, \rho_2]$ uniquely identify Φ and $\rho_2^{\downarrow 2L}$ up to the action of $\text{SO}(3)$ on $(\Phi, \rho_2^{\downarrow 2L})$ and up to chirality.*

The proof of Theorem 4.1 is at the end of this section.

Remark 4.2. Theorem 4.1 identifies the structure up to an overall global rotation and reflection. This is an unavoidable ambiguity in cryo-EM, and therefore not a drawback of this particular result. For clarity, we include Lemma 4.3 below which formalizes the ambiguity.

Lemma 4.3. *Let Φ, ρ_1, ρ_2 and $\tilde{\Phi}, \rho_1, \tilde{\rho}_2$ both satisfy Assumption 2.5. Assume they differ from each other by a global rotation and possibly chirality, that is, there exist $S \in \text{SO}(3)$ and $\epsilon \in \{0, 1\}$ such that for all $\mathbf{x} \in \mathbb{R}^3$ and $R \in \text{SO}(3)$ it holds*

$$\tilde{\Phi}(\mathbf{x}) = \Phi(J^\epsilon S \mathbf{x}) \quad \text{and} \quad \tilde{\rho}_2(R) = \rho_2(J^\epsilon S R J^\epsilon), \quad (4.1)$$

where $J = \text{diag}(1, 1, -1)$. Then we have

$$m_1[\Phi, \rho_1] = m_1[\tilde{\Phi}, \rho_1], \quad m_2[\Phi, \rho_1] = m_2[\tilde{\Phi}, \rho_1], \quad \text{and} \quad m_2[\Phi, \rho_2] = m_2[\tilde{\Phi}, \tilde{\rho}_2]. \quad (4.2)$$

See Appendix B for a proof of Lemma 4.3.

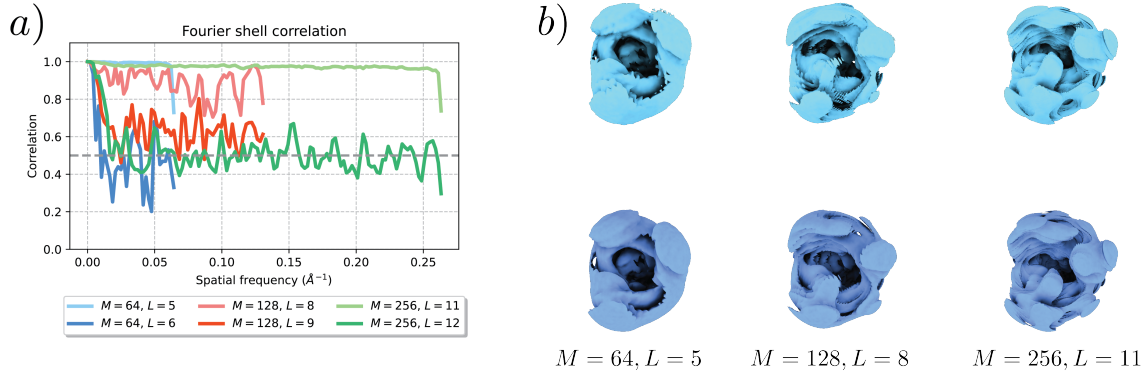


Figure 3.1: Result of running Alg. 1 on the dataset EMD-2660 [59] from the online electron microscopy data bank [34]. (a) FSC curves for the reconstructins, for different values of bandlimit L and image size M . (b) Reconstructions (top) and ground truth structures (bottom).

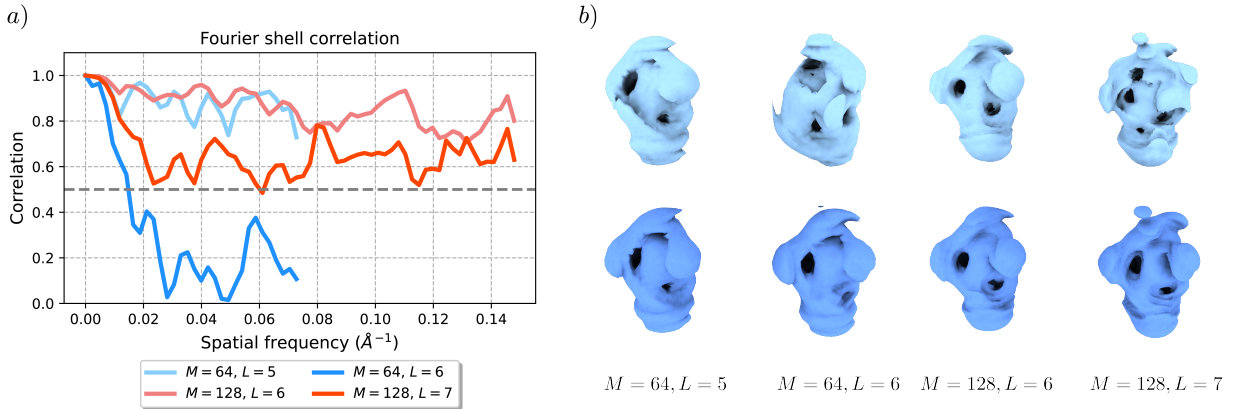


Figure 3.2: Result of running Alg. 1 on the dataset EMD-32743 [27] from the online electron microscopy data bank [34]. (a) FSC curves for the reconstructins, for different values of bandlimit L and image size M . (b) Reconstructions (top) and ground truth structures (bottom).

Remark 4.4. Our proof strategy for Theorem 4.1 goes by induction on L . The verification of the base case $L = 3$ requires computer assistance. In particular, we use floating-point arithmetic and pseudo-random numbers for the base case. Although such calculations are standard practice in the field of computational algebra, they fall short of providing a completely rigorous proof; thus the proof of Theorem 4.1 is labeled as a “Computational Proof” below. All other steps in the argument are rigorous.

Our induction step for Theorem 4.1 relies on the following lemmas. The lemmas imply that certain linear systems arising in the proof generically have a unique solution. For

convenience, we denote by $B_p = (B_{p,u})_{-p \leq u \leq p}$ the vector in $\mathbb{C}^{2p+1}$.

Lemma 4.5. *Let $P \geq 2L$. If the expansion coefficients $(B_p : 2 \leq p \leq 2L - 2, p \text{ is even})$ are fixed, then the affine-linear map*

$$B_{2L} \mapsto (\mathcal{B}_{L,L}^n : 0 \leq n \leq L, n \equiv L \pmod{2}) \quad (4.3)$$

is injective.

Lemma 4.6. *Let $L \geq 4$. If the expansion coefficients $(B_p : 2 \leq p \leq 2L - 2, p \text{ is even})$ are Zariski-generic, then the horizontal concatenation of the matrices*

$$(\mathcal{B}_{L,\ell'}^n : 0 \leq \ell' < L, n \equiv L \equiv \ell' \pmod{2}) \quad (4.4)$$

has full column rank.

See Appendix B for proofs of Lemmas 4.5 and 4.6. We are now ready to prove our main theoretical result.

Computational Proof of Theorem 4.1. Let Φ , ρ_1 , and ρ_2 satisfy Assumption 2.5. Suppose $\tilde{\Phi}$ and $\tilde{\rho}_2$ are another structure and distribution with expansions

$$\hat{\Phi}(r, \theta, \varphi) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \tilde{A}_{\ell}^m(r) Y_{\ell}^m(\theta, \varphi), \quad r \in [0, r_{\max}], \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi), \quad (4.5)$$

and

$$\tilde{\rho}_2(R) dR = \sum_{\substack{p=0 \\ p \text{ even}}}^P \sum_{u=-p}^p \tilde{B}_{p,u} U_{u0}^p(R) dR, \quad (4.6)$$

such that there is a matching of population moments:

$$m_1[\Phi, \rho_1] = m_1[\tilde{\Phi}, \rho_1], \quad m_2[\Phi, \rho_1] = m_2[\tilde{\Phi}, \rho_1], \quad m_2[\Phi, \rho_2] = m_2[\tilde{\Phi}, \tilde{\rho}_2]. \quad (4.7)$$

Our goal is to prove that $(\tilde{\Phi}, \tilde{\rho}_2^{\downarrow 2L})$ equals $(\Phi, \rho_2^{\downarrow 2L})$ up to rotation and possibly chirality. As in Lemma 4.3, this precisely means there exist $S \in \text{SO}(3)$ and $\epsilon \in \{0, 1\}$ such that $\tilde{\Phi}(\mathbf{x}) = \Phi(J^{\epsilon} S \mathbf{x})$ and $\tilde{\rho}_2^{\downarrow 2L}(R) = \rho_2^{\downarrow 2L}(J^{\epsilon} S R J^{\epsilon})$. Notice that $\tilde{\Phi}(\mathbf{x}) = \Phi(J^{\epsilon} S \mathbf{x})$ is equivalent to

$$\tilde{A}_{\ell}(r) = A_{\ell}(r) U^{\ell}(J^{\epsilon} S), \quad \text{for all } 0 \leq \ell \leq L, \quad (4.8)$$

because, writing $\mathbf{x} = (r, \theta, \varphi)$ in spherical coordinates, we have

$$\hat{\Phi}(J^{\epsilon} S \mathbf{x}) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} A_{\ell m}(r) \sum_{n=-\ell}^{\ell} U_{mn}^{\ell}(J^{\epsilon} S) Y_{\ell}^n(\theta, \varphi) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} [A_{\ell}(r) U^{\ell}(J^{\epsilon} S)]_m Y_{\ell}^m(\theta, \varphi),$$

where $A_{\ell}(r)$ denotes the row-vector-valued function evaluated at the radial frequency r , as defined in (2.12).

Toward this goal, we first use the condition $m_2[\Phi, \rho_1] = m_2[\tilde{\Phi}, \rho_1]$. By [30], it implies

$$\tilde{A}_\ell(r) = A_\ell(r) Q_\ell^H O_\ell^T Q_\ell, \quad \text{for all } 0 \leq \ell \leq L, \quad (4.9)$$

where $O_\ell \in \mathbb{R}^{(2\ell+1) \times (2\ell+1)}$ are some unknown real-valued orthogonal matrices and $Q_\ell \in \mathbb{C}^{(2\ell+1) \times (2\ell+1)}$ are the complex-valued unitary matrices defined in (2.2), which represent the unitary transforms from the complex to the real spherical harmonics basis (cf. (3.5)). Next, we use $m_1[\Phi, \rho_1] = m_1[\tilde{\Phi}, \rho_1]$. This implies $\tilde{A}_0(r) = A_0(r)$, again by [30]. Since $Q_0^H = 1$ it follows that

$$O_0 = 1, \quad (4.10)$$

where $O_0 \in \mathbb{R}$ is the 1×1 orthogonal matrix in (4.9).

Next, we claim that by the symmetry in (4.8), it is without loss of generality to assume

$$O_1 = I \in \mathbb{R}^{3 \times 3}. \quad (4.11)$$

Indeed, if $O_1 \neq I$ there exist $S \in \text{SO}(3)$ and $\epsilon \in \{0, 1\}$ such that $Q_1^H U^1(J^\epsilon S) Q_1 = O_1$ (see [16, pg. 324]). Then the replacement $\Phi(\mathbf{x}) \leftarrow \Phi(J^\epsilon S \mathbf{x})$ reduces us to $O_1 = I$ (cf. (4.9)). Note that assuming $O_1 = I$ kills rotational and chiral ambiguities of the problem, i.e., the goal becomes to prove that these equalities hold exactly, $\tilde{\Phi} = \Phi$ and $\tilde{\rho}_2 = \rho_2$. Thus with (4.11), we want

$$O_\ell = I \in \mathbb{R}^{(2\ell+1) \times (2\ell+1)}, \quad \text{for all } 0 \leq \ell \leq L, \text{ and} \quad (4.12)$$

$$\tilde{B}_{p,u} = B_{p,u}, \quad \text{for all } 0 \leq p \leq 2L \text{ with } p \text{ even and } -p \leq u \leq p. \quad (4.13)$$

To show (4.12) and (4.13), we turn to the condition $m_2[\Phi, \rho_2] = m_2[\tilde{\Phi}, \tilde{\rho}_2]$. First by (A.15),

$$m_2[\Phi, \rho_2](r, \varphi, r', \varphi') = \sum_{n=-L}^L e^{in(\varphi-\varphi')} \sum_{\ell=0}^L \sum_{\ell'=0}^L A_\ell(r) \mathcal{B}_{\ell,\ell'}^n(A_{\ell'}(r'))^H. \quad (4.14)$$

Combining with (4.9),

$$m_2[\tilde{\Phi}, \tilde{\rho}_2](r, \varphi, r', \varphi') = \sum_{n=-L}^L e^{in(\varphi-\varphi')} \sum_{\ell=0}^L \sum_{\ell'=0}^L A_\ell(r) Q_\ell^H O_\ell^T Q_\ell \tilde{\mathcal{B}}_{\ell,\ell'}^n Q_{\ell'}^H O_{\ell'} Q_{\ell'}(A_{\ell'}(r'))^H. \quad (4.15)$$

Equating (4.14) and (4.15) and using orthonormality of the Fourier modes deduce

$$\sum_{\ell=0}^L \sum_{\ell'=0}^L A_\ell(r) \mathcal{B}_{\ell,\ell'}^n(A_{\ell'}(r'))^H = \sum_{\ell=0}^L \sum_{\ell'=0}^L A_\ell(r) Q_\ell^H O_\ell^T Q_\ell \tilde{\mathcal{B}}_{\ell,\ell'}^n Q_{\ell'}^H O_{\ell'} Q_{\ell'}(A_{\ell'}(r'))^H \quad (4.16)$$

for each n satisfying $-L \leq n \leq L$. By the assumed linear independence of the radial functions A_ℓ^m (i.e., the second item of Assumption 2.5), this implies

$$\mathcal{B}_{\ell,\ell'}^n = Q_\ell^H O_\ell^T Q_\ell \tilde{\mathcal{B}}_{\ell,\ell'}^n Q_{\ell'}^H O_{\ell'} Q_{\ell'} \quad (4.17)$$

for each (n, ℓ, ℓ') with $-L \leq n \leq L$ and $0 \leq \ell, \ell' \leq L$. Here $\tilde{\mathcal{B}}_{\ell, \ell'}^n$ depends on $\tilde{B}_p$ as $\mathcal{B}_{\ell, \ell'}^n$ does on B_p .

In the rest of the proof, our strategy is to leverage (4.17) for (n, ℓ, ℓ') in an appropriate order to establish (4.12) and (4.13) by induction on L . Precisely, we induct on the following:

Take Assumption 2.5. Then (4.17) for $-L \leq n \leq L$ and $0 \leq \ell, \ell' \leq L$, together with (4.10) and (4.11), imply $O_\ell = I$ for $0 \leq \ell \leq L$ and $\tilde{B}_p = B_p$ for $p = 0, 2, \dots, 2L$. (4.18)

The induction amounts to showing that a polynomial system in O_ℓ and $\tilde{B}_p$ has a unique solution.

The base case of the induction is $L = 3$. By (2.10), (4.10) and (4.11), we know $B_0 = \tilde{B}_0$, $O_0 = 1$ and $O_1 = I$. We wish to show $\tilde{B}_2 = B_2$, $\tilde{B}_4 = B_4$, $\tilde{B}_6 = B_6$, $O_2 = I$ and $O_3 = I$. We will use equations (4.17) suitably rearranged, together with orthogonality constraints:

$$\begin{cases} Q_\ell^H O_\ell Q_\ell \mathcal{B}_{\ell, \ell'}^n Q_{\ell'}^H O_{\ell'}^T Q_{\ell'} = \tilde{\mathcal{B}}_{\ell, \ell'}^n \\ O_2 O_2^T = O_2^T O_2 = I \\ O_3 O_3^T = O_3^T O_3 = I, \end{cases} \quad (4.19)$$

where in the first line $(n, \ell, \ell') \in \{(1, 1, 1), (0, 2, 0), (0, 2, 2), (1, 3, 1), (1, 3, 3), (3, 3, 3)\}$. Dropping reality constraints on O_ℓ and $\tilde{B}_p$, we view (4.19) as a parameterized polynomial system over $\mathbb{C}$: the variables are $(\tilde{B}_2, \tilde{B}_4, \tilde{B}_6, O_2, O_3) \in \mathbb{C}^{101}$, the parameters are $(B_2, B_4, B_6) \in \mathbb{C}^{27}$, and there are $(3 \times 3) + (5 \times 1) + (5 \times 5) + (5 \times 5) + (7 \times 3) + (7 \times 7) + (7 \times 7) + (5 \times 5) + (5 \times 5) + (7 \times 7) + (7 \times 7) = 331$ equations. Note the equations are affine-quadratic or linear in the variables and affine-linear in the parameters. By general properties of parameterized polynomial systems over $\mathbb{C}$, there exists a nonempty Zariski-open subset $\mathcal{U} \subseteq \mathbb{C}^{27}$ such that for $(B_2, B_4, B_6) \in \mathcal{U}$ the solution set to (4.19) has the same “type” of irreducible decomposition in the sense of [54, Theorem A.14.10]. Therefore, if we show on a randomly-generated instance of (B_2, B_4, B_6) that the polynomial system (4.19) has a unique solution over $\mathbb{C}$, then with probability 1 the system Zariski-generically has a unique solution over $\mathbb{C}$, which then must be the trivial solution $\tilde{B}_p = B_p$ for $p = 2, 4, 6$ and $O_\ell = I$ for $\ell = 2, 3$. Checking polynomial systems on random instances is a standard approach in computational algebra; that said, Remark 4.4 applies.

Here we perform the check using the numerical homotopy continuation and computer algebra software [13, 25]. We generate (B_2, B_4, B_6) using a random number generator. The system (4.19) is too big to directly input into the software, so we break up the computation. First, using the top line of (4.19) when $(n, \ell, \ell') = (1, 1, 1)$ it follows by a linear solve or by Lemma 4.5 that $\tilde{B}_2 = B_2$. Secondly, we take $(n, \ell, \ell') = (0, 2, 0)$, which gives the equation $Q_2^H O_2 Q_2 \mathcal{B}_{2,0}^0 = \mathcal{B}_{2,0}^0$ or

$$O_2 Q_2 \mathcal{B}_{2,0}^0 = Q_2 \mathcal{B}_{2,0}^0, \quad (4.20)$$

i.e., O_2 fixes a known vector. We find that the vector is non-isotropic, i.e., $(Q_2 \mathcal{B}_{2,0}^0)^T (Q_2 \mathcal{B}_{2,0}^0) \neq 0$, so we can extend the vector suitably scaled to a complex orthogonal matrix, i.e., find $\tilde{O}_2^T \in \mathcal{O}(5, \mathbb{C})$ and $\lambda \in \mathbb{C}$ such that the leftmost column of $\tilde{O}_2^T$ is $\lambda Q_2 \mathcal{B}_{2,0}^0$. Then $\tilde{O}_2^T e_1 = \lambda Q_2 \mathcal{B}_{2,0}^0$

or $\lambda \tilde{O}_2 Q_2 \mathcal{B}_{2,0}^0 = e_1$, where e_1 is the first standard basis, and (4.20) can be rewritten as $(\tilde{O}_2 O_2 \tilde{O}_2^\top) \lambda \tilde{O}_2 Q_2 \mathcal{B}_{2,0}^0 = \lambda \tilde{O}_2 Q_2 \mathcal{B}_{2,0}^0$ or

$$(\tilde{O}_2 O_2 \tilde{O}_2^\top) e_1 = e_1.$$

Using $(\tilde{O}_2 O_2 \tilde{O}_2^\top)^\top (\tilde{O}_2 O_2 \tilde{O}_2^\top) = I$, it follows that $\tilde{O}_2 O_2 \tilde{O}_2^\top = 1 \oplus \tilde{\tilde{o}}_2$ for some $\tilde{\tilde{o}}_2 \in \mathcal{O}(4, \mathbb{C})$, i.e., $\tilde{O}_2 O_2 \tilde{O}_2^\top$ is block-diagonal. Next, consider $(n, \ell, \ell') = (0, 2, 2), (2, 2, 2)$ in (4.19):

$$Q_2^H (1 \oplus \tilde{\tilde{o}}_2) Q_2 \mathcal{B}_{2,2}^n Q_2^H (1 \oplus \tilde{\tilde{o}}_2)^\top Q_2 = \tilde{\mathcal{B}}_{2,2}^n, \quad (4.21)$$

for $n = 0, 2$. The right-hand side depends affine-linearly on $\tilde{B}_4$ and on no other unknowns. Therefore, we can linearly eliminate $\tilde{B}_4$ from (4.21). We find $5 \times 5 - 9 = 16$ affine-quadratic equations in $\tilde{\tilde{o}}_2$ for $n = 0, 2$. We solve the resulting polynomial system of these 36 equations together with $\tilde{\tilde{o}}_2 \tilde{\tilde{o}}_2^\top = \tilde{\tilde{o}}_2^\top \tilde{\tilde{o}}_2 = I$ in the variables $\tilde{\tilde{o}}_2 \in \mathbb{C}^{16}$ using the software [13]. Four isolated multiplicity-1 solutions are computed: $\tilde{\tilde{o}}_2^{(1)}, \tilde{\tilde{o}}_2^{(2)}, \tilde{\tilde{o}}_2^{(3)}, \tilde{\tilde{o}}_2^{(4)} \in \mathbb{C}^{16}$. We return to (4.21), substitute in these solutions for $\tilde{\tilde{o}}_2$ and linearly solve for $\tilde{B}_4 \in \mathbb{C}^9$. Corresponding to each $\tilde{\tilde{o}}_2^{(i)}$ we find a unique solution $\tilde{B}_4^{(i)} \in \mathbb{C}^9$. Next up, use (4.19) with $(n, \ell, \ell') = (1, 3, 1)$:

$$O_3 Q_3 \mathcal{B}_{3,1}^1 = Q_3 \tilde{\mathcal{B}}_{3,1}^1. \quad (4.22)$$

Left-multiplying each side by its transpose yields

$$(\mathcal{B}_{3,1}^1)^\top Q_3^\top Q_3 \mathcal{B}_{3,1}^1 = (\tilde{\mathcal{B}}_{3,1}^1)^\top Q_3^\top Q_3 \tilde{\mathcal{B}}_{3,1}^1. \quad (4.23)$$

To evaluate $\tilde{\mathcal{B}}_{3,1}^1$ we plug in the possible values for $\tilde{B}_4$, namely $\tilde{B}_4^{(1)}, \tilde{B}_4^{(2)}, \tilde{B}_4^{(3)}, \tilde{B}_4^{(4)}$, and find that only one of these satisfies (4.23). It is the value that equals B_4 , and corresponds to $\tilde{\tilde{o}}_2 = I$ or $\tilde{O}_2 O_2 \tilde{O}_2^\top = I$. We conclude $\tilde{B}_4 = B_4$ and $O_2 = I$. Next, we reuse (4.22): since $\tilde{\mathcal{B}}_{3,1}^1 = \mathcal{B}_{3,1}^1$, it says that O_3 fixes a known 7×3 matrix. Similarly to how we utilized (4.20), here we find $\tilde{O}_3 \in \mathcal{O}(7, \mathbb{C})$ such that the first three columns of $\tilde{O}_3^\top$ span the column space of $Q_1 \mathcal{B}_{3,\infty}^\infty$ and $\tilde{O}_3 O_3 \tilde{O}_3^\top = I_3 \oplus \tilde{\tilde{o}}_3$ for some $\tilde{\tilde{o}}_3 \in \mathcal{O}(4, \mathbb{C})$ where I_3 denotes the 3×3 identity matrix. Then, consider (4.19) with $(n, \ell, \ell') = (1, 3, 3), (3, 3, 3)$:

$$Q_3^H (I_3 \oplus \tilde{\tilde{o}}_3) Q_3 \mathcal{B}_{3,3}^n Q_3^H (I_3 \oplus \tilde{\tilde{o}}_3)^\top Q_3 = \tilde{\mathcal{B}}_{3,3}^n, \quad (4.24)$$

for $n = 1, 3$. Similarly to how we dealt with (4.21), we linearly eliminate $\tilde{B}_6$ from (4.24). We find $7 \times 7 - 13 = 36$ affine-quadratic equations in $\tilde{\tilde{o}}_3$ for $n = 1, 3$. The polynomial system of these 72 equations with $\tilde{\tilde{o}}_3 \tilde{\tilde{o}}_3^\top = \tilde{\tilde{o}}_3^\top \tilde{\tilde{o}}_3 = I$ in variables $\tilde{\tilde{o}}_3 \in \mathbb{C}^{16}$ is solved using [13]. The software computes a unique solution, which is multiplicity-1: $\tilde{\tilde{o}}_3 = I$. Hence $\tilde{O}_3 O_3 \tilde{O}_3^\top = I$, or $O_3 = I$. We return to (4.24) which now reads $\mathcal{B}_{3,3}^n = \tilde{\mathcal{B}}_{3,3}^n$ for $n = 1, 3$. Linearly solving for $\tilde{B}_6$ or using Lemma 4.5, deduce $\tilde{B}_6 = B_6$. This wraps up the base case of (4.18).

Let us now turn to the induction step. Thus assume $L \geq 4$, and that (4.18) has been shown for $L - 1$. We wish to show (4.18) for L . By the inductive hypothesis, we know $O_\ell = I$ for $0 \leq \ell \leq L - 1$ and $B_p = \tilde{B}_p$ for $p = 0, 2, \dots, 2(L - 1)$. We want $O_L = I$ and $\tilde{B}_{2L} = B_{2L}$. We will rely on two lemmas concerning the dependence of $\mathcal{B}_{\ell,\ell'}^n$ on B_p , Lemmas 4.5 and 4.6.

Consider (4.17) for (n, L, ℓ') satisfying $0 \leq \ell' < L, n \equiv L \equiv \ell' \pmod{2}$. For these triples $\mathcal{B}_{L,\ell'}^n = \tilde{\mathcal{B}}_{L,\ell'}^n$, because $(\mathcal{B}_{L,\ell'}^n)$ depend only on (B_p) with $p \leq 2L - 2$, $(\tilde{\mathcal{B}}_{L,\ell'}^n)$ depend only on $(\tilde{B}_p)$ with $p \leq 2L - 2$ in the same way, and $B_p = \tilde{B}_p$ for $p \leq 2L - 2$. Therefore (4.17) reads

$$\mathcal{B}_{L,\ell'}^n = Q_L^H O_L^T Q_L \mathcal{B}_{L,\ell'}^n,$$

where we used $O_{\ell'} = I$. So left multiplication by $Q_L^H O_L^T Q_L$ fixes the concatenation of matrices:

$$(\mathcal{B}_{L,\ell'}^n : 0 \leq \ell' < L, n \equiv L \equiv \ell' \pmod{2}).$$

By Lemma 4.6, the concatenation has full column rank. Thus $Q_L^H O_L^T Q_L = I$, or $O_L = I$.

Next, consider (4.17) for (n, L, L) where $0 \leq n \leq L$ and $n \equiv L \pmod{2}$. These read:

$$\mathcal{B}_{L,L}^n = \tilde{\mathcal{B}}_{L,L}^n,$$

using $O_L = I$. As $B_p = \tilde{B}_p$ for $p \leq 2L - 2$, Lemma 4.5 gives $\tilde{B}_{2L} = B_{2L}$. This finishes the induction step for (4.18).

Putting everything together, we conclude that under Assumption 2.5, the moments $m_1[\Phi, \rho_1], m_2[\Phi, \rho_1], m_2[\Phi, \rho_2]$ uniquely identify Φ and $\rho_2^{\downarrow 2L}$, up to rotation and chirality. $\square$

Remark 4.7. We stress that although the details of Theorem 4.1 are tailored to the image formation model of cryo-EM, we envision that similar results will hold for related imaging techniques like XFEL. Similarly, the Algorithm 1 is likely adaptable to other settings.

5 Conclusion

In this paper, we introduced a new moment-based algorithm for 3-D reconstruction in cryo-electron microscopy that fuses information from multiple datasets. By using a dataset with uniformly distributed viewing angles and another dataset with non-uniformly distributed viewing angles, we showed that the 3-D reconstruction problem in cryo-EM becomes easier. In particular, we proved a stronger uniqueness guarantee for reconstruction based on the first and second-order moments than previously available. Building on this theory, we devised an efficient numerical scheme using alternating optimization, where each subproblem admits a closed-form solver. Numerical experiments show accurate molecular reconstruction is achieved by the framework up to specified bandlimits. The paper opens the door to data fusion in cryo-EM and related imaging modalities.

The approach has limitations that we aim to address in future work. One direction is to develop statistical tests to check whether experimentally collected datasets have uniform distributions of viewing angles. Another goal is to handle higher bandlimits L , through appropriate modification of Algorithm 1. One more avenue is to adapt the proposed framework to datasets in X-ray free-electron laser imaging, where similar inverse problems arise.

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A Proofs for Section 2

This appendix derives the basis expansion of the rotation distribution in Section 2.2 and the analytic moment expressions in Section 2.3.

A.1 Derivation of basis expansion of rotation distribution

We assume that the rotational distribution has density $\rho(R)$ with respect to the uniform distribution dR . By Peter-Weyl theorem [16, Thm. 8.13], any square integrable function $\rho(R)$ can be expanded as

$$\rho(R) = \sum_{p=0}^P \sum_{u,v=-p}^p B_{p,u,v} U_{uv}^p(R),$$

as $P \rightarrow \infty$, where $R \in \text{SO}(3)$ and $U_{uv}^p(R)$ is the (u, v) -th entry of the Wigner U-matrix $U^p(R) \in \mathbb{C}^{(2p+1) \times (2p+1)}$ [16, Eq. 9.41].

We next show how distributions invariant to in-plane rotations can be expressed as in (2.8). According to the definition (2.6), we have

$$\sum_{p=0}^P \sum_{u,v=-p}^p B_{p,u,v} U_{uv}^p(R) = \sum_{p=0}^P \sum_{u,v=-p}^p B_{p,u,v} U_{uv}^p(Rz(\alpha)) = \sum_{p=0}^P \sum_{u,v=-p}^p B_{p,u,v} \left(U^p(R) U^p(z(\alpha)) \right)_{uv},$$

where we use the homomorphism property of group representations [16, Page 344] for Wigner U-matrices in the last inequality and

$$U^p(z(\alpha)) = \text{diag}(e^{-ip\alpha}, e^{-i(p-1)\alpha}, \dots, e^{ip\alpha}).$$

This gives

$$\sum_{p=0}^P \sum_{u,v=-p}^p B_{p,u,v} U_{uv}^p(R) = \sum_{p=0}^P \sum_{u,v=-p}^p B_{p,u,v} U_{uv}^p(R) e^{iv\alpha},$$

which implies $B_{p,u,v} = 0$ for $v \neq 0$. Thus, $\rho(R)$ admits the expansion

$$\rho(R) = \sum_{p=0}^P \sum_{u=-p}^p B_{p,u,0} U_{u0}^p(R).$$

For simplicity, we drop the zero subscript and write

$$\rho(R)dR = \sum_{p=0}^P \sum_{u=-p}^p B_{p,u} U_{u0}^p(R) dR, \quad (\text{A.1})$$

where dR denotes the uniform distribution and $U_{u0}^p(R)$ can explicitly be written as in (2.9).

To ensure that the density $\rho(R)$ is a real valued function, we impose the conjugate symmetry condition on its expansion coefficients. Specifically, we require

$$\sum_{p=0}^P \sum_{u=-p}^p B_{p,u} U_{u0}^p(R) = \sum_{p=0}^P \sum_{u=-p}^p \overline{B_{p,u} U_{u0}^p(R)}.$$

Using the identity $\overline{U_{u0}^p(R)} = (-1)^u U_{-u,0}^p(R)$ [11, Page 68], we obtain the symmetry condition

$$\overline{B_{p,u}} = (-1)^u B_{p,-u}. \quad (\text{A.2})$$

Additionally, since $\rho(R)$ is a probability density over $\text{SO}(3)$, it must integrate to one. Using the fact that $U_{00}^0(R) = 1$ for any $R \in \text{SO}(3)$, and applying the orthogonality of the matrix elements $\{U_{uv}^p(R) : p \geq 0, -p \leq u, v \leq p\}$, we get

$$B_{0,0} = \int_{\text{SO}(3)} \sum_{p=0}^P \sum_{u=-p}^p B_{p,u} U_{u0}^p(R) dR = \int_{\text{SO}(3)} \rho(R) dR = 1.$$

Finally, for chirality invariance, we deduce from the definition (2.7) that

$$\sum_{p=0}^P \sum_{u=-p}^p B_{p,u} U_{u0}^p(R) = \sum_{p=0}^P \sum_{u=-p}^p B_{p,u} U_{u0}^p(JRJ).$$

Using the identity

$$U_{u0}^p(JRJ) = (-1)^p U_{u0}^p(R),$$

we obtain

$$B_{p,u} = 0, \quad \text{for all odd } p.$$

A.2 Derivation of moment expressions

Using the Fourier slice theorem [41] (c.f. its Cartesian version in (1.3)) and the property of Wigner U-matrices [16, Eqn. 9.49], we can write the Fourier transform of the projection images in (1.2) in spherical coordinates (r, θ, φ) as

$$\begin{aligned} \hat{I}_R(r, \varphi) &= (R^\top \cdot \hat{\Phi})(r, \frac{\pi}{2}, \varphi) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} A_{\ell}^m(r) \sum_{n=-\ell}^{\ell} U_{mn}^{\ell}(R) Y_{\ell}^n(\frac{\pi}{2}, \varphi) + \hat{\varepsilon}(r, \varphi) \\ &= \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \sum_{n=-\ell}^{\ell} A_{\ell}^m(r) U_{mn}^{\ell}(R) e^{in\varphi} N_{\ell}^n + \hat{\varepsilon}(r, \varphi), \end{aligned} \quad (\text{A.3})$$

where $\widehat{\varepsilon}$ denotes the Fourier transform of the noise term, and in the last equality we use the explicit expressions for spherical harmonics [16, Eqn. 4.36] and define

$$N_\ell^n = \sqrt{\frac{2\ell+1}{4\pi}} \sqrt{\frac{(\ell-n)!}{(\ell+n)!}} P_\ell^n(0),$$

where $P_\ell^n(x)$ denotes the associated Legendre polynomials. Notably, N_ℓ^n is always real-valued and, by the symmetry property of $P_\ell^n(x)$ [16, Page 53], we have

$$\begin{aligned} N_\ell^{-n} &= \sqrt{\frac{2\ell+1}{4\pi}} \sqrt{\frac{(\ell+n)!}{(\ell-n)!}} P_\ell^{-n}(0) = \sqrt{\frac{2\ell+1}{4\pi}} \sqrt{\frac{(\ell+n)!}{(\ell-n)!}} \cdot (-1)^n \cdot \frac{(\ell-n)!}{(\ell+n)!} P_\ell^n(0) \\ &= (-1)^n N_\ell^n. \end{aligned} \quad (\text{A.4})$$

A.2.1 First-order moment

We first derive the formula for the first-order population moment. Using (A.3) and applying the zero mean property of Gaussian noise (see also 1.7), we can write the first-order population moment as

$$\begin{aligned} m_1(r, \varphi) &= \mathbb{E}_\varepsilon \left[\int_{\text{SO}(3)} \widehat{I}_R(r, \varphi) \rho(R) dR \right] \\ &= \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \sum_{n=-\ell}^{\ell} A_\ell^m(r) e^{in\varphi} N_\ell^n \sum_{p=0}^P \sum_{u=-p}^p B_{p,u} \int_{\text{SO}(3)} U_{u0}^p(R) U_{mn}^\ell(R) dR. \end{aligned} \quad (\text{A.5})$$

Note that Wigner U-matrices satisfy the orthogonality property [16, Eq. 9.32]

$$\int \overline{U_{mn}^\ell(R)} U_{uv}^p(R) dR = \frac{1}{2\ell+1} \mathbf{1}_{\ell=p} \mathbf{1}_{u=m} \mathbf{1}_{v=n},$$

and the symmetry property [11, Page 68]

$$\overline{U_{mn}^\ell(R)} = (-1)^{m+n} U_{-m,-n}^\ell(R),$$

we can simplify (A.5) to

$$m_1(r, \varphi) = \sum_{\ell=0}^{\min\{L,P\}} \sum_{m=-\ell}^{\ell} (-1)^m \frac{1}{2\ell+1} N_\ell^0 \cdot A_\ell^m(r) B_{\ell,-m}, \quad (\text{A.6})$$

where

$$N_\ell^0 = \sqrt{\frac{2\ell+1}{4\pi}} P_\ell^0(0)$$

with

$$P_\ell^0(0) = \begin{cases} (-1)^{\ell/2} \cdot \frac{(\ell-1)!!}{\ell!!}, & \text{if } \ell \text{ is even,} \\ 0, & \text{if } \ell \text{ is odd.} \end{cases}$$

The sum in (A.6) ranges over integer indices (ℓ, m) such that

$$0 \leq \ell \leq \min\{L, P\}, \quad \ell \text{ is even,} \quad -\ell \leq m \leq \ell,$$

which gives

$$m_1(r, \varphi) = \sum_{\substack{\ell=0, \\ \ell \text{ even}}}^{\min\{L, P\}} \sum_{m=-\ell}^{\ell} (-1)^m \frac{N_\ell^0 A_\ell^m(r)}{2\ell+1} B_{\ell, -m} = \sum_{\substack{\ell=0, \\ \ell \text{ even}}}^{\min\{L, P\}} \sum_{m=-\ell}^{\ell} \frac{N_\ell^0 A_\ell^m(r)}{2\ell+1} \overline{B_{\ell, m}}. \quad (\text{A.7})$$

Thus, we can write the first-order moment in the following matrix form:

$$m_1(r, \varphi) = \sum_{\ell=0}^{\min\{L, P\}} \sum_{m=-\ell}^{\ell} \frac{N_\ell^0 \mathbf{1}_{\{\ell \text{ is even}\}}}{2\ell+1} A_{\ell m}(r) \overline{B_{\ell, m}} = \sum_{0 \leq \ell \leq \min\{L, P\}} A_\ell(r) (\mathfrak{B}_\ell)^H, \quad (\text{A.8})$$

where the terms

$$A_\ell(r) \in \mathbb{C}^{1 \times (2\ell+1)} \quad \text{and} \quad \mathfrak{B}_\ell \in \mathbb{C}^{1 \times (2\ell+1)}$$

for $0 \leq \ell \leq \min\{L, P\}$ are defined by

$$(A_\ell(r))_m = A_\ell^m(r), \quad (\text{A.9})$$

$$\mathcal{N}_\ell^0 = N_\ell^0 \cdot \mathbf{1}_{\{\ell \equiv 0 \pmod{2}\}} \cdot \mathbf{1}_{\{\ell \geq 0\}}, \quad (\text{A.10})$$

$$(\mathfrak{B}_\ell)_m = \frac{N_\ell^0 B_{\ell, m}}{2\ell+1}, \quad (\text{A.11})$$

for $-\ell \leq m \leq \ell$. Here $A_\ell(r)$ and $\mathcal{N}_\ell^0$ are the same as defined in (2.12) and (2.14), respectively.

A.2.2 Second-order moment

We now derive the formula for the second-order population moment and prove Proposition 2.2. Using (A.3) and noting that the noise term $\hat{\varepsilon}$ is independent from the signal term, we can write the second order population moment from (1.8) as

$$\begin{aligned} m_2(r, \varphi, r', \varphi') &= \int_{\text{SO}(3)} \left(\sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \sum_{n=-\ell}^{\ell} A_\ell^m(r) U_{mn}^\ell(R) e^{in\varphi} N_\ell^n \right) \left(\sum_{\ell'=0}^L \sum_{m'=-\ell'}^{\ell'} \sum_{n'=-\ell'}^{\ell'} \overline{A_{\ell'}^{m'}(r')} \overline{U_{m'n'}^{\ell'}(R)} e^{-in'\varphi'} N_{\ell'}^{n'} \right) \\ &\quad \times \left(\sum_{p=0}^P \sum_{u=-p}^p B_{p,u} U_{u0}^p(R) \right) dR. \end{aligned}$$

The product of two Wigner matrix entries can be expressed as a linear combination of Wigner matrix entries [16, Eqn. 9.64],

$$U_{mn}^\ell(R)U_{m'n'}^{\ell'}(R) = \sum_{\ell''=|\ell-\ell'|}^{\ell+\ell'} \mathcal{C}_{\ell''}(\ell, \ell', m, m', n, n') U_{m+m', n+n'}^{\ell''}(R), \quad (\text{A.12})$$

where

$$\mathcal{C}_{\ell''}(\ell, \ell', m, m', n, n') = C(\ell, m; \ell', m' | \ell'', m+m') C(\ell, n; \ell', n' | \ell'', n+n'), \quad (\text{A.13})$$

is the product of two Clebsch-Gordan coefficients. This product is nonzero only if (ℓ, ℓ', ℓ'') satisfy the triangle inequalities $|\ell - \ell'| \leq \ell'' \leq \ell + \ell'$. Note that we also use the fact that the Clebsch-Gordan coefficient $C(\ell, m; \ell', m' | \ell'', m'')$ is nonzero only if $m'' = m + m'$ to simplify the formula in (A.12) [12, Eqn 2.41]. Using the identity $\overline{U_{b,c}^a(R)} = U_{-b,-c}^a(R)$, we obtain

$$\begin{aligned} & \int_{\text{SO}(3)} U_{mn}^\ell(R) \overline{U_{m'n'}^{\ell'}(R)} U_{u0}^p(R) dR \\ &= \int_{\text{SO}(3)} (-1)^{m'+n'} U_{mn}^\ell(R) U_{-m', -n'}^{\ell'}(R) U_{u0}^p(R) dR \\ &= \int_{\text{SO}(3)} (-1)^{m'+n'} \sum_{\ell''=|\ell-\ell'|}^{\ell+\ell'} \mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n') U_{m-m', n-n'}^{\ell''}(R) U_{u0}^p(R) dR \\ &= \int_{\text{SO}(3)} (-1)^{m+n} \sum_{\ell''=|\ell-\ell'|}^{\ell+\ell'} \mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n') \overline{U_{-m+m', -n+n'}^{\ell''}(R)} U_{u0}^p(R) dR \\ &= (-1)^{m+n} \sum_{\ell''=|\ell-\ell'|}^{\ell+\ell'} \mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n') \frac{1}{2\ell''+1} \mathbf{1}_{\ell''=p} \mathbf{1}_{m'-m=u} \mathbf{1}_{n=n'}. \end{aligned}$$

It follows that

$$\begin{aligned} & m_2(r, \varphi, r', \varphi') \\ &= \sum_{\ell=0}^L \sum_{\ell'=0}^L \sum_{m=-\ell}^{\ell} \sum_{m'=-\ell'}^{\ell'} \sum_{n=-\ell}^{\ell} \sum_{n'=-\ell'}^{\ell'} e^{in\varphi} e^{-in'\varphi'} N_\ell^n N_{\ell'}^{n'} A_\ell^m(r) \overline{A_{\ell'}^{m'}(r')} \\ &\times \sum_{p=0}^P \sum_{u=-p}^p \sum_{\ell''=|\ell-\ell'|}^{\ell+\ell'} B_{p,u} (-1)^{m+n} \frac{\mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n')}{2\ell''+1} \mathbf{1}_{p=\ell''} \mathbf{1}_{u=m'-m} \mathbf{1}_{n=n'}. \end{aligned}$$

Rearranging the order of summations gives

$$\begin{aligned}
& m_2(r, \varphi, r', \varphi') \\
&= \sum_{n=-L}^L \sum_{n'=-L}^L \sum_{\ell=|n|}^L \sum_{\ell'=|n'|}^L \sum_{m=-\ell}^{\ell} \sum_{m'=-\ell'}^{\ell'} e^{in\varphi} e^{-in'\varphi'} N_{\ell}^n N_{\ell'}^{n'} A_{\ell}^m(r) \overline{A_{\ell'}^{m'}(r')} \\
&\times \sum_{p=0}^P \sum_{u=-p}^p \sum_{\ell''=|\ell-\ell'|}^{\ell+\ell'} B_{p,u}(-1)^{m+n} \frac{\mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n')}{2\ell''+1} \mathbf{1}_{p=\ell''} \mathbf{1}_{u=m'-m} \mathbf{1}_{n=n'}.
\end{aligned}$$

This further gives

$$\begin{aligned}
& m_2(r, \varphi, r', \varphi') \\
&= \sum_{n=-L}^L \sum_{\ell=|n|}^L \sum_{\ell'=|n|}^L e^{in(\varphi-\varphi')} N_{\ell}^n N_{\ell'}^n \\
&\times \sum_{m=-\ell}^{\ell} \sum_{m'=-\ell'}^{\ell'} \sum_{p=0}^P \sum_{u=-p}^p \sum_{\ell''=|\ell-\ell'|}^{\ell+\ell'} A_{\ell}^m(r) \overline{A_{\ell'}^{m'}(r')} B_{p,u}(-1)^{m+n} \mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n) \frac{\mathbf{1}_{p=\ell''} \mathbf{1}_{u=m'-m}}{2\ell''+1} \\
&= \sum_{n=-L}^L \sum_{\ell=|n|}^L \sum_{\ell'=|n|}^L e^{in(\varphi-\varphi')} N_{\ell}^n N_{\ell'}^n \\
&\times \sum_{\ell''=|\ell-\ell'|}^{\min\{\ell+\ell', P\}} \sum_{\substack{-\ell \leq m \leq \ell \\ -\ell' \leq m' \leq \ell' \\ |m'-m| \leq \ell''}} (-1)^{m+n} \mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n) \frac{A_{\ell}^m(r) \overline{A_{\ell'}^{m'}(r')}}{2\ell''+1} B_{\ell'', m'-m}.
\end{aligned}$$

According to [22, Lemma D.1], we get $N_{\ell}^n = 0$ for odd $\ell + n$, which further implies

$$\begin{aligned}
& m_2(r, \varphi, r', \varphi') \\
&= \sum_{n=-L}^L e^{in(\varphi-\varphi')} \sum_{\substack{|n| \leq \ell \leq L \\ \ell \equiv n \pmod{2}}} \sum_{\substack{|n| \leq \ell' \leq L \\ \ell' \equiv n \pmod{2}}} N_{\ell}^n N_{\ell'}^n \\
&\sum_{\ell''=|\ell-\ell'|}^{\min\{\ell+\ell', P\}} \sum_{\substack{-\ell \leq m \leq \ell \\ -\ell' \leq m' \leq \ell' \\ |m'-m| \leq \ell''}} (-1)^{m+n} \mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n) \frac{A_{\ell}^m(r) \overline{A_{\ell'}^{m'}(r')}}{2\ell''+1} B_{\ell'', m'-m}. \quad (\text{A.14})
\end{aligned}$$

As anticipated by the fact that the rotational distribution ρ is in-plane uniform, the second-order moment depends only on $\varphi - \varphi'$ as shown above. Rearranging the sum and using the

notation $\mathcal{N}_\ell^n$ as defined in (2.14) gives

$$\begin{aligned}
m_2(r, \varphi, r', \varphi') &= \sum_{n=-L}^L e^{in(\varphi-\varphi')} \sum_{\ell=0}^L \sum_{\ell'=0}^L \mathcal{N}_\ell^n \mathcal{N}_{\ell'}^n \\
&\sum_{\substack{-\ell \leq m \leq \ell \\ -\ell' \leq m' \leq \ell'}} \sum_{\ell''=\max\{|m-m'|, |\ell-\ell'|\}}^{\min\{\ell+\ell', P\}} (-1)^{m+n} \frac{\mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n)}{2\ell''+1} A_\ell^m(r) B_{\ell'', m'-m} \overline{A_{\ell'}^{m'}(r')}.
\end{aligned}$$

Thus, we can write the second-order moment more concisely in the following matrix form:

$$m_2(r, \varphi, r', \varphi') = \sum_{n=-L}^L e^{in(\varphi-\varphi')} \sum_{0 \leq \ell \leq L} \sum_{0 \leq \ell' \leq L} A_\ell(r) \mathcal{B}_{\ell, \ell'}^n (A_{\ell'}(r'))^H, \quad (\text{A.15})$$

where the terms

$$A_\ell(r) \in \mathbb{C}^{1 \times (2\ell+1)}, \quad \mathcal{B}_{\ell, \ell'}^n \in \mathbb{C}^{(2\ell+1) \times (2\ell'+1)},$$

for $\ell, \ell' \in \{0, \dots, L\}$, $n \in \{-L, \dots, L\}$ are defined as (2.12) and (2.13).

In addition, we show the Hermitian property of $(\mathcal{B}_{\ell', \ell}^n)_{m', m}$ for potential application in the paper. By the symmetry property of Clebsh-Gordan coefficients [12, Eqn. 2.47],

$$\begin{aligned}
\mathcal{C}_{\ell''}(\ell', \ell, m', -m, n, -n) &= C(\ell', m'; \ell, -m | \ell'', -m+m') \cdot C(\ell', n; \ell, -n | \ell'', 0) \\
&= C(\ell, -m; \ell', m' | \ell'', -m+m') \cdot C(\ell, -n; \ell', n | \ell'', 0) \\
&= C(\ell, m; \ell', -m' | \ell'', m-m') \cdot C(\ell, n; \ell', -n | \ell'', 0) \\
&= \mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n).
\end{aligned} \quad (\text{A.16})$$

Applying (A.2) and (A.4), we obtain

$$\begin{aligned}
\overline{(\mathcal{B}_{\ell', \ell}^n)_{m', m}} &= \sum_{\ell''=\max\{|m-m'|, |\ell-\ell'|\}}^{\min\{\ell+\ell', P\}} (-1)^{m'+n} \mathcal{N}_{\ell'}^n \mathcal{N}_\ell^n \frac{\mathcal{C}_{\ell''}(\ell', \ell, m', -m, n, -n)}{2\ell''+1} \overline{B_{\ell'', m-m'}} \\
&= \sum_{\ell''=\max\{|m-m'|, |\ell-\ell'|\}}^{\min\{\ell+\ell', P\}} (-1)^{m'+n} \mathcal{N}_{\ell'}^n \mathcal{N}_\ell^n \frac{\mathcal{C}_{\ell''}(\ell', \ell, m', -m, n, -n)}{2\ell''+1} (-1)^{m-m'} B_{\ell'', -m+m'} \\
&= \sum_{\ell''=\max\{|m-m'|, |\ell-\ell'|\}}^{\min\{\ell+\ell', P\}} (-1)^{m+n} \mathcal{N}_\ell^n \mathcal{N}_{\ell'}^n \frac{\mathcal{C}_{\ell''}(\ell, \ell', m, -m', n, -n)}{2\ell''+1} B_{\ell'', -m+m'} \\
&= (\mathcal{B}_{\ell, \ell'}^n)_{m, m'}.
\end{aligned}$$

Hence, we get

$$\mathcal{B}_{\ell, \ell'}^n = (\mathcal{B}_{\ell', \ell}^n)^H.$$

Also, according to the definition (2.14), we know $\mathcal{N}_\ell^n \mathcal{N}_{\ell'}^n = 0$ whenever $\ell \not\equiv \ell' \pmod{2}$, which implies for any $-L \leq n \leq L$,

$$\mathcal{B}_{\ell, \ell'}^n = 0,$$

whenever $\ell \not\equiv \ell' \pmod{2}$.

B Proofs for Section 4

B.1 Proof of Lemma 4.3

In fact, we claim that the distributions of tomographic projection images I_R with noise removed (that is, (1.2) without the $\varepsilon(x, y)$ term) as generated by (Φ, ρ_1) and (Φ, ρ_2) match the distributions generated by $(\tilde{\Phi}, \rho_1)$ and $(\tilde{\Phi}, \tilde{\rho}_2)$, respectively. Consequently, $m_k[\Phi, \rho_1] = m_k[\tilde{\Phi}, \rho_1]$ and $m_k[\Phi, \rho_2] = m_k[\tilde{\Phi}, \tilde{\rho}_2]$ for all $k \geq 1$. In particular, (4.2) holds as wanted.

Let us justify this claim. First compare (Φ, ρ_2) and $(\tilde{\Phi}, \tilde{\rho}_2)$. Under $(\tilde{\Phi}, \tilde{\rho}_2)$, the distribution of noiseless tomographic projection images draws with density $\tilde{\rho}_2(R) = \rho_2(J^\epsilon S R J^\epsilon)$,

$$\int_{-\infty}^{\infty} (R^\top \cdot \tilde{\Phi})(\mathbf{x}) dx_3 = \int_{-\infty}^{\infty} \Phi(J^\epsilon S R \mathbf{x}) dx_3 = \int_{-\infty}^{\infty} \Phi(J^\epsilon S R J^\epsilon J^\epsilon \mathbf{x}) dx_3 = \int_{-\infty}^{\infty} \Phi(J^\epsilon S R J^\epsilon \mathbf{x}) dx_3. \quad (\text{B.1})$$

Here the last equality in (B.1) is by a change of variable replacing x_3 by $(-1)^\epsilon x_3$. Meanwhile, under (Φ, ρ_2) the distribution of noiseless tomographic projection images draws

$$\int_{-\infty}^{\infty} \Phi(J^\epsilon S R J^\epsilon \mathbf{x}) dx_3, \quad (\text{B.2})$$

with density $\rho_2(J^\epsilon S R J^\epsilon)$. The distributions are identical, as claimed. Comparing (Φ, ρ_1) and $(\tilde{\Phi}, \rho_1)$, the image distributions likewise are the same, because $\rho_1(R) = \rho_1(J^\epsilon S R J^\epsilon)$ since ρ_1 is the uniform distribution. This justifies our claim and finishes the proof of the lemma.

B.2 Proof of Lemma 4.5

In fact, we will show that the affine-linear map

$$B_{2L} \mapsto \mathcal{B}_{L,L}^{\underline{n}} \quad (\text{B.3})$$

is injective, where $\underline{n}$ is the element of $\{0, 1\}$ satisfying $\underline{n} \equiv L \pmod{2}$. By (2.13), the linear part of (B.3) (i.e., dropping an additive constant depending only B_p with $p \leq 2L - 2$) reads:

$$B_{2L} \mapsto (\mathcal{B}_{L,L}^{\underline{n}})_{m,m'} = (-1)^{m+\underline{n}} \mathcal{N}_L^{\underline{n}} \mathcal{N}_L^{\underline{n}} \frac{\mathcal{C}_{2L}(L, L, m, -m', \underline{n}, -\underline{n})}{4L+1} B_{2L, m'-m}.$$

By Lemmas B.1 and B.2, the coefficient of $B_{2L, m'-m}$ is nonzero for all $-L \leq m, m' \leq L$. Clearly, (B.3) is injective and so is (4.3).

B.3 Proof of Lemma 4.6

Since $(\mathcal{B}_{L,\ell'}^n)$ depend polynomially on (B_p) and full column rank of (4.4) is a Zariski-open condition [54], it is enough to exhibit a single instance of B_p such that (4.4) has full column rank. Furthermore, in producing such an instance we can temporarily drop the conjugate symmetry condition in (2.10) on (B_p) , because this condition defines a dense subset with respect to the complex Zariski topology. Moreover, we can also temporarily drop the normalization condition in (2.10), because $(\mathcal{B}_{L,\ell'}^n)$ depend homogeneously on (B_p) .

Now, let us produce a desired instance. If $L = 4$ or $L = 5$, randomly generate (B_p) and verify that (4.4) has rank $2L + 1$ (to be fully rigorous, in exact arithmetic in a finite extension of $\mathbb{Q}$ using the formulas for $\mathcal{C}_{\ell''}(\ell, \ell', m, m', n, n')$ and $\mathcal{N}_{\ell}^n$ in the proofs of Lemmas B.1 and B.2). Meanwhile, for $L \geq 6$ choose (B_p) as follows:

$$B_{p,u} = \begin{cases} \text{nonzero and sufficiently large in magnitude if } (p, u) = (2L - 2, 2) \\ \text{nonzero and sufficiently small in magnitude if } (p, u) = (2L - 4, -2L + 7) \\ 0 \text{ for all other } (p, u). \end{cases}$$

In (4.4), consider the submatrix $[\mathcal{B}_{L,L-2}^n | \mathcal{B}_{L,L-4}^n] \in \mathbb{C}^{(2L+1) \times (4L-10)}$ where $\underline{n} \in \{0, 1\}$ such that $\underline{n} \equiv L \pmod{2}$. The rows of the submatrix are indexed by $m \in \{-L, -L + 1, \dots, L\}$. The columns of its first and second block are indexed by $m' \in \{-(L-2), -(L-2) + 1, \dots, L-2\}$ and $m' \in \{-(L-4), -(L-4) + 1, \dots, L-4\}$ respectively. With the above choice of (B_p) , the block $\mathcal{B}_{L,L-2}^n$ has support contained in the main diagonal where $m' - m = -(L-2) - (-L) = 2$ and the off-diagonal where $m' - m = -(L-2) - (L-5) = -2L + 7$ by (2.13). Moreover, the entries in the main diagonal are nonzero by Lemmas B.1 and B.2, and larger in magnitude than the entries in the off-diagonal. Meanwhile, $\mathcal{B}_{L,L-4}^n$ is supported on its off-diagonal where $m' - m = -(L-4) - (L-3) = -2L + 7$, and these entries are nonzero by Lemmas B.1 and B.2. It follows that the leftmost $(2L + 1) \times (2L + 1)$ submatrix of $[\mathcal{B}_{L,L-2}^n | \mathcal{B}_{L,L-4}^n]$ is columnwise diagonally dominant, and therefore of full rank. This finishes the case $L \geq 6$.

B.4 Nonvanishing of constants

Lemma B.1. *Let integers $\ell, \ell', m, m', n, n'$ satisfy $\ell \geq |m|, |n|$ and $\ell' \geq |m'|, |n'|$. Then the constant $\mathcal{C}_{\ell+\ell'}(\ell, \ell', m, m', n, n')$ (2.16) is nonzero.*

Proof. For all septuples of integers satisfying

$$\ell, \ell', \ell'' \geq 0, \quad |\ell - \ell'| \leq \ell'' \leq \ell + \ell', \quad \ell \geq |m|, |n| \quad \text{and} \quad \ell' \geq |m'|, |n'|,$$

by (2.16) $\mathcal{C}_{\ell''}(\ell, \ell', m, m', n, n') = C(\ell, m; \ell', m' | \ell'', m + m') C(\ell, n; \ell', n' | \ell'', n + n')$; by [12]

$$\begin{aligned} & C(\ell, m; \ell', m' | \ell'', m + m') \\ &= \sqrt{\frac{(2\ell'' + 1)(\ell + \ell' - \ell'')!(\ell + \ell'' - \ell')!(\ell' + \ell'' - \ell)!}{(\ell + \ell' + \ell'' + 1)!}} \\ & \times \sqrt{(\ell - m)!(\ell + m)!(\ell' - m')!(\ell' + m')!(\ell'' - m'')!(\ell'' + m'')!} \\ & \times \sum_k \frac{(-1)^k}{k!(\ell + \ell' - \ell'' - k)!(\ell - m - k)!(\ell' + m' - k)!(\ell'' - \ell' + m + k)!(\ell'' - \ell - m' + k)!} \end{aligned}$$

where the summation is over all integers k such that the argument of every factorial is nonnegative; and [12] the analogous formula for $C(\ell, n; \ell', n' | \ell'', n + n')$ holds. In the special case $\ell'' = \ell + \ell'$, the summations over k collapse to one term where $k = 0$. It follows $C(\ell, m; \ell', m' | \ell + \ell', m + m') \neq 0$ and $C(\ell, n; \ell', n' | \ell + \ell', n + n') \neq 0$, hence $\mathcal{C}_{\ell+\ell'}(\ell, \ell', m, m', n, n') \neq 0$. $\square$

Lemma B.2. *Let integers ℓ and n satisfy $\ell \geq |n|$ and $\ell \equiv n \pmod{2}$. Then the constant $\mathcal{N}_\ell^n$ (2.14) is nonzero.*

Proof. Under the stated conditions on ℓ and n , by (2.14)

$$\mathcal{N}_\ell^n = \sqrt{\frac{2\ell + 1}{4\pi}} \sqrt{\frac{(\ell - n)!}{(\ell + n)!}} P_\ell^n(0),$$

where $P_\ell^n(x)$ denotes an associated Legendre polynomial. By [16], for $-1 \leq x \leq 1$ it holds

$$P_\ell^n(x) = \frac{(-1)^n}{2^\ell \ell!} (1 - x^2)^{n/2} \frac{d^{\ell+n}}{dx^{\ell+n}} (x^2 - 1)^\ell.$$

The coefficient of $x^{\ell+n}$ in $(x^2 - 1)^\ell$ is $\binom{\ell}{(\ell+n)/2} (-1)^{(\ell-n)/2}$, therefore $\frac{d^{\ell+n}}{dx^{\ell+n}} (x^2 - 1)^\ell|_{x=0} = (\ell + n)! \binom{\ell}{(\ell+n)/2} (-1)^{(\ell-n)/2}$. Hence $P_\ell^n(0) = \frac{(-1)^{(\ell+n)/2}}{2^\ell \ell!} (\ell + n)! \binom{\ell}{(\ell+n)/2}$ and

$$\mathcal{N}_\ell^n = \frac{(-1)^{(\ell+n)/2}}{2^\ell} \sqrt{\frac{2\ell + 1}{4\pi}} \frac{\sqrt{(\ell + n)! (\ell - n)!}}{\left(\frac{\ell+n}{2}\right)! \left(\frac{\ell-n}{2}\right)!}. \quad (\text{B.4})$$

In particular, $\mathcal{N}_\ell^n$ is nonzero as claimed. $\square$